Barriers to industrial growth and long-run productive efficiency: the role of policy-induced misallocation *

Abhishek Deshwal† Tianyu Luo‡

November 5, 2026

Please click here for the most recent version

Abstract

We study how government-created distortions in upstream coal markets affect capacity investment and productivity in India's electricity industry. The government coal monopoly administratively allocates subsidized coal primarily to inefficient state-owned plants while forcing efficient private generators to purchase coal at market prices often 1.5-2 times higher. This unequal access distorts not only the allocation of electricity production today, but also incentives to invest in new generating capacity. Using plant-level data for 2006-2017 and detailed administrative records of coal supply contracts, we estimate a dynamic oligopoly model of capacity investment for coal-fired generators. We use the estimated model to quantify how eliminating distortions would change investment and the long-run productivity of the electricity sector.

*Thanks to Gautam Gowrisankaran, Jeffrey Shrader, Pietro Tebaldi, Sebastian Otero, and many seminar participants for helpful comments. Thanks to Varun Ramamurthy, Ria Vashishtha, and Anandamayi Bajaj for excellent Research Assistance. All errors are own own.

†School of International and Public Affairs, Columbia University. ad3884@columbia.edu

‡Department of Economics, Columbia University tl3088@columbia.edu

1 Introduction

Production misallocation, and the resulting aggregate productivity losses are widely recognized as an important determinant of cross-country differences in productivity. Recent literature has established that significant productivity gains can be realized through the reallocation of resources from less productive to more productive firms (Hsieh and Klenow (2009); Busso et al. (2013)). Beyond static effects, which much of this literature has focused on, distortions also shape investment, including how much capacity is built, the productivity and timing of investment, and therefore the growth trajectory of an industry or economy. There is little empirical evidence on this dynamic channel.

In this paper, we study how administrative coal allocation and pricing policies in India shaped investment and productivity in the electricity generation industry. We focus on how a government-created distortion in upstream coal markets systematically advantages inefficient producers while discouraging efficient producers from investing. Because fuel costs are central to the profitability of coal power plants, this distortion affects not only who generates electricity today, but also who invests in new capacity and when. Investment timing matters in this context because the value of investment changed significantly over the period – severe electricity shortages in the early 2010s and excess but severely misallocated capacity later in the decade. We model private firms' capacity investment decisions and quantify the impact of alternative coal policies on output and cost during the 2006 to 2021 period.

The electricity industry in India provides a nice setting to examine these questions for several reasons. First, a 2003 Reform incentivized large entry of private firms in electricity generation, formerly dominated by public firms. Yet electricity shortages persisted for years after the reform, in part because private investment lagged growth in demand. Second, the industry exhibits substantial productivity dispersion, with modern private plants achieving heat rates (energy consumption per unit of electricity generated) of 2200 kcal/kWh compared to 2400-2500 kcal/kWh for public plants of similar vintage. Third, a policy distortion introduced a large tax wedge on the efficient private producers: the administrative allocation of coal by Coal India Limited at regulated prices set significantly below market rates. While public-sector generators received substantial allocations through long-term fuel supply agreements, private generators relied more heavily on expensive spot-market coal or imports, at premiums often 2-3 times higher.

The cost disparity between private and public generators lowers their incentive to invest in additional capacity, but the direction and size of the effect of removing it is ex ante unclear. Higher fuel costs push generators up the supply curve, so they generate less often and earn lower profits on the capacity they own. But removing the cost differences has two opposing effects on investment. On the one hand, eliminating the distortion would let efficient private generators compete with public generators on equal terms, raising the return on new investment. On the other hand, as more investment enters the market, total generation capacity becomes less scarce relative to demand. Consequently, electricity prices fall, which lowers the return on each additional unit invested and offsets the gain from uniform coal pricing. Even setting scarcity aside, capacity investment is costly and lumpy, and changes in fuel costs may not move expected profits enough (relative to the investment costs) to change investment decisions. How strongly the coal distortion impacts investment is therefore an empirical question.

To answer this, we develop a dynamic oligopoly model of investment in generation capacity. We estimate the per-period profit function, a key input to the dynamic model, with a random forest. Then, we estimate investment cost parameters using observed investment decisions; specifically, we choose parameters to match model-implied investment moments to their data analogs. Finally, we solve for equilibria under alternative policy environments.

We model private firms choosing capacity investment in an infinite-horizon game. When making those decisions, firms account for their current capacity, access to (subsidized) domestic coal, as well as the expected path of coal policy that determines how much domestic coal will be available to new investments. New investments become operational in the following period. Because public firms have objectives other than maximizing profits, we treat public capacity as evolving exogenously.

We confront the curse of dimensionality, a common challenge for dynamic oligopoly models, by adopting the Approximate Belief Oligopoly Equilibrium (ABOE) of Gowrisankaran et al. (2025). In a standard Markov-perfect Equilibrium, each firm would track every other firm's states, e.g. their capacities and domestic coal access. With two states per firm and ten firms per market, we would have twenty states, making the model intractable. Under ABOE, each firm tracks only its own state and a small set of aggregate market states, chosen to summarize important determinants of profit. And firms recognize that their own investment decision can impact future aggregate

states. This substantially reduces the state space, while also allowing for strategic interactions between firms.

We obtain the per-period profit function by estimating a random forest with rich data. While it is standard in industrial organization to take a more microfounded approach, estimating demand and supply functions and then computing equilibrium profit, doing so is difficult in our setting. Modeling profit would require us to both model PPA auctions as well as generators' competition in producing electricity, moderated by system operators. To focus on dynamic investment decisions, we instead construct realized profits directly, using detailed PPA price, quantity, and coal contract data. We use a random forest to estimate realized profit as a function of aggregate and firm-specific states, which accommodates nonlinearities and interactions between states. The estimated profit function achieves an out-of-sample R^2 of 0.83.

Using the estimated profit function, we estimate investment cost parameters by full-solution GMM. To prepare for this, we estimate how domestic coal access evolves across periods, i.e., coal policy path; we find that domestic coal access for new investment is most favorable for period-2 investments (which mature in period 3), and considerably worse in the other two periods. Then, we estimate investment cost parameters by matching model-implied investment moments to their data counterparts. Probability of investing informs the fixed cost of investment. The level of investment informs the marginal cost of investment. And dispersion in investment by firms in similar states informs the variance of investment cost shocks. Our estimates for all three parameters are imprecise; calibrating marginal investment cost from Elliott (2025), we obtain fairly precise estimates of the other two parameters, which we regard as preferred estimates. The preferred estimates fit the main investment moments quite well, but they underpredict the concentration of investment among observations with large capacity.

Using the preferred estimates, we conduct two sets of counterfactuals. The first removes differences in domestic coal access in two ways. In one, we estimate public firms' domestic coal access over time from data, and apply the same law of motion to private firms. In the other, private firms are guaranteed access to coal no lower than the public average. The second set of counterfactuals asks whether high-powered incentives to invest early can help alleviate severe electricity shortages observed in the early years of the analysis period. Outages are costly and temporary subsidies can elicit large investment responses (Zwick and Mahon (2017)).

To measure the consequences of each counterfactual for output, costs, and shortages, we combine the simulated investment paths with a model of the electricity supply curve. For each counterfactual, the investment model gives private capacity and domestic coal access by market and period. We add public capacity, which we treat as exogenous and take from the data. We then construct a monthly supply curve for each regional market, ordering generators by marginal cost, and clear it against inelastic demand. This yields generation by plant, the marginal cost of supply, generation costs, and unmet demand. New private capacity is assigned the average heat rate of private plants in its market and their average transport cost in its market and period. We value unmet demand at INR 5 per kWh, roughly the average residential tariff during this period.

We find that removing public-private differences in domestic coal access has a small impact on aggregate investment. The first implementation, applying to private firms the domestic coal access transition of public firms, raises total investment by 168 MW relative to the factual simulation, while shifting some investment from the first and the third period to the second, where domestic coal access is most generous. The second implementation, providing private firms no lower domestic coal access than the average over public firms, yields 1087 MW less total investment than the factual. To make sense of the negative effect, note that the second implementation provides a coal policy that is equally good over time, whereas the factual coal policy is considerably better in period 2 than in the other two periods. Firms, aware that the most attractive policy is temporary, respond by investing more in period 2. Under a more uniform coal policy, firms are more likely to wait.

In terms of welfare, the first set of counterfactuals led to gains in productive efficiency of 7.4 and 10.3 billion Rupees per year, from 2010 to 2017. Those gains do not come from the mechanical reduction in cost when receiving domestic coal access. Rather, they come from displacing high-cost generators with low-cost ones. They have limited effect on lowering shortages.

The second set of counterfactuals, offering 100 percent domestic coal access to period-1 investments, and none afterward, reallocates investment forward substantially. Specifically, period 1 investment is 9000 MW higher relative to the factual, although end-of-period capacity is similar. In terms of welfare, we realize productive efficiency gains of 9.4 billion Rupees per year. The magnitude may appear small relative to the large impact on investment timing, but it can be explained by severe shortage

in early years, where the accelerated investment comes online; because demand exceeds supply, all generating capacity is deployed, and no gains accrue from replacing high-cost generators with low-cost ones. By contrast, we realize much larger gains from lowering shortages relative to the previous set of counterfactuals. Accelerating investment relieves outages sooner, and realizes gains of 26.3 billion Rupees per year.

Related Literature First, we contribute to the literature on misallocation and productivity dispersion. The foundational work by Hsieh and Klenow (2009) and Busso et al. (2013) demonstrated that cross-sectional differences in marginal revenue products across producers within narrowly defined industries can account for substantial aggregate productivity differences between countries. This insight has spawned a large literature attempting to identify the specific sources of such misallocation, with mixed empirical success (Restuccia and Rogerson (2017)). Studies examining particular distortions—financial frictions (Banerjee and Duflo (2014)), trade barriers (Melitz and Redding (2014)), regulation (Olley and Pakes (1996)) or land market institutions (Besley and Ghatak (2010)) have typically found effects that are orders of magnitude smaller than the cross or within-country differences documented in the foundational work.

Our paper addresses three empirical challenges noted by this literature. First, we address productivity measurement challenges by using a direct, physical measure of power plant efficiency—the plant heat rate—which sidesteps traditional production function estimation problems. Second, we directly observe and measure the distortion using detailed administrative data on coal supply contracts, allocation mechanisms, and pricing, allowing us to quantify the precise magnitude of producer-level distortions rather than inferring it indirectly. Third, the policy-created nature of the distortion provides an important source of variation for identification and establishing credible counterfactuals.

Our focus on investment connects the paper to works on the dynamic consequences of misallocation. Firms may respond to distortions through entry and exit (Melitz (2003)), technology investment (Midrigan and Xu (2014)), or capital adjustment (Asker et al. (2014))—and can generate impacts that are significantly larger than static effects (Da-Rocha et al. (2016); Peters (2020)). However, papers identifying specific, directly observed causes of dynamic misallocation remain scarce. Most studies rely on indirect approaches estimating misallocation from observed marginal revenue product differences, requiring strong assumptions about production functions

and market structure.

While recent work has made important progress in understanding reallocation mechanisms through dynamic processes, our approach differs in key ways. For instance, Collard-Wexler and De Loecker (2015) demonstrate how technological change drives reallocation in the steel industry, focusing on the role of process innovation in long-run productivity growth. Asker et al. (2014) show how apparent misallocation can arise endogenously from optimal dynamic adjustment to shocks, emphasizing that some resource misallocation may reflect efficient responses to adjustment costs rather than market failures. In contrast, we examine misallocation explicitly created by government policy, administrative coal allocation mechanisms that create artificial coal scarcity and cost disparities between firms, preventing efficient reallocation even in the presence or absence of natural adjustment frictions. Using detailed microdata and a structural model, we quantify how this distortion affects capacity investment.

Lastly, our paper relates to empirical models of dynamic games, starting with Ericson and Pakes (1995) and Pakes and McGuire (1994). We use ABOE to reduce the state space. ABOE differs from Oblivious Equilibrium (Weintraub et al. (2008)), by allowing firms to recognize they can influence aggregate states. And ABOE differs from Moment-based Equilibrium (Ifrach and Weintraub (2017)) by allowing all firms, not just dominant firms, to recognize this influence. ABOE seems suitable in our setting where each market consists of around ten private firms, none clearly dominant. We combine ABOE with a fast algorithm for solving for investment decisions by Gowrisankaran and Schmidt-Dengler (2025). We estimate our model by full solution, rather than a Conditional Choice Probability estimator. Because coal-policy path changes over time, we cannot pool observed investment decisions across periods to estimate the conditional choice probability function precisely.

2 Background and Data

2.1 Private sector entry in India's electricity generation industry

After independence in 1947, the Indian power sector was mostly publicly owned and run for more than 40 years. A sweeping economic liberalization in the early 1990s opened power generation to private firms, and the Indian Government solicited investment, including from foreign companies. For the power sector specifically, this period of liberalization was a failure. The deregulation of entry in generation reduced state

control over generation, but retained the state's primary role in the transmission and distribution of electricity. As deregulation was not paired with any deeper structural reform, potential private entrants still faced monopsony buyers, the State Electricity Boards, in every state, and were therefore reluctant to invest (Mathavan (2008)).

Figure 1 shows the thermal generation capacity in the Indian electricity Industry from 2001 to 2019, and capacity by ownership (public which comprises state and central governments, and private). Excluding very old power plants that retired before 2006, the privately-owned share of capacity is nearly zero ($<$ 1 GW) until 2001, and is slow-growing thereafter.

The failure of this generation of private projects and power rationing built up pressure for deeper reforms. The Electricity Act of 2003 undertook structural reforms that recognized the natural monopoly nature of much of the power sector (Kumar and Chatterjee (2012)). Under these laws, the State Electricity Boards were separated into component parts, for generation, transmission, and distribution. Independent regulators were established to rule on power contracts and tariffs. Markets for power, though initially a small share of the sector, provided an outside option for private entrants to sell power (Ryan (2021)). The National Tariff Policy (2006), issued under Section 63 of the Electricity Act (2003), mandated that state procurement of power must be done through a competitive bidding process (Ministry of Power (2006)). The auction results are formalized in a power purchase agreement (PPA), a contract between the supplier and procuring state utility or utilities that is written at the price set by the auction and reviewed and approved by the electricity regulator.

Several features of this procurement process made the industry attractive to private power sellers: the existence of Electricity Regulatory Commissions to approve contracts and arbitrate disputes; the principle of contracts being revealed at auction, which may lower contract prices and increase the transparency of price setting; the deference of the institutions now involved to the market process of setting prices; the large amount of power they are supplying, which may make them difficult to replace in the short term; and finally, the fact that the buy side consists of consortia of state bidders for large projects, which diversifies the political risk faced by sellers (Ryan (2020)).

There was a large wave of private investment in power in response to the deeper structural reforms in the electricity sector and under the new bidding rules. The rules took effect in 2006 but the new projects built under these rules typically had

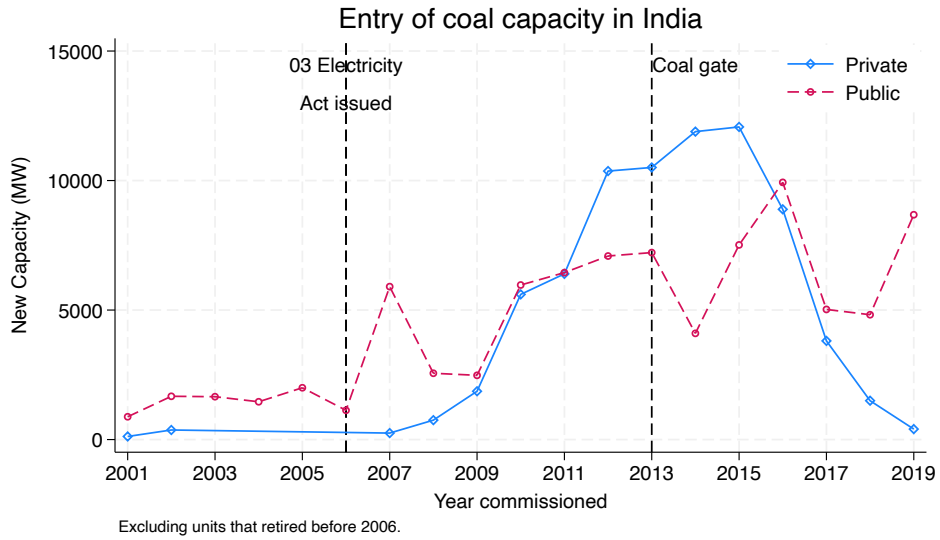
a 4-5 year lag, meaning that they came online in 2010. Returning to Figure 1, we see a rapid increase in private generation capacity during this period, as shown by the blue curve, such that by 2017 private generation capacity was over 35 percent of the total capacity, greater than that owned by either the states or the Central Government. This dramatic transformation positioned private generators as major players in India's electricity sector, with many demonstrating superior operational efficiency compared to their public counterparts.

Notably, since 2018, there has been little to no increase in private capacity. Considering the 4-year lag to build, very few projects announced since 2013-14 have proceeded to completion with multiple cancellations, defaults or postponements. The timing of this cessation in private investment coincides with the emergence of widespread financial distress in the private thermal power sector. By 2017-18, 40 GW of upcoming and installed thermal capacity was classified as stressed assets (assets that have lost or are nearing to lose their ability to service their debt obligations) with INR 1.74 lakh crore in outstanding dues to creditors, affecting major private developers including Lanco Infratech, KSK Energy, JayPee Power, GMR Energy and Coastal Energen (Standing Committee on Energy (2018)).[1].

The abrupt cessation of private investment after 2017, despite continued demand growth and the apparent success of the regulatory framework in attracting private capital, presents a puzzle. While demand saturation and renewable energy competition may partially explain this pattern, the timing (renewables weren't a competitive threat in 2014) and severity of the private sector's retreat suggests additional institutional factors may have fundamentally altered the investment landscape during this period. We argue in the next section that the institutional design governing the supply and pricing of coal for private generators was the leading cause.

[1]The Parliamentary Standing Committee identified coal supply issues as one of the top cause of stress. The report noted: "*One of the major reasons for stress was cited by the Promoters of the stressed was the lack of adequate amount of Fuel Supply Agreement. The Committee, therefore, had discussions with the Ministry of coal and tried to have the true picture of availability coal in the country especially for the power sector.*" The Ministry of Power testified that "*The total 34 projects of 40,000 MW capacity, 17 projects are affected because of coal linkage – may be partial or full nonavailability of coal linkage. The total capacity of 17 projects was around 15,200 MW*"

Figure 1. Power plant entry in India over time



2.2 Coal Market Distortions in India's Power Sector

India's power sector is characterized by significant institutional distortions in coal markets that create inefficiencies in generation and impact generator profits. At the heart of these distortions lies Coal India Limited's (CIL) near monopoly over coal production and distribution, coupled with a complex regulatory framework that systematically disadvantages private generators. The core design and evolution of coal policies during 2007-17 created differential fuel access between public and private generators, with direct implications for industry-wide productivity.

CIL generates 84 percent of India's domestic coal supply, with the remaining produced by Singareni Collieries Company Limited (SCCL) and captive mines. While CIL holds monopoly power, it doesn't exercise monopoly pricing. Instead, coal is primarily sold through notified prices, which are administratively determined, grade-specific rates that implicitly carry subsidies. These prices are set to balance CIL's profitability with the government's objective of keeping coal prices low for end-users. Only 10-20 percent of total production is sold through the domestic spot market or e-auctions, where the premium prices reflect coal's scarcity value (Ministry of Coal (2014)). The economic significance of these institutional arrangements becomes apparent when considering that coal typically accounts for 60-70% of variable generation costs, making fuel access the primary determinant of plant competitiveness.

The real market for coal in India is essentially access to coal contracts or fuel supply agreements (FSAs) that specify supply guarantees at notified prices, but with explicit priority given to the power sector and further preferential treatment to public sector users. Those lacking secure access to domestic coal must rely on more expensive imports, creating a two-tier market where producer-level economic outcomes depend on administrative fuel access rather than productivity, directly affecting both current profits and future investment decisions.

2.2.1 Evolution of coal allocation policies

The institutional framework governing coal allocation evolved in ways that systematically disadvantaged private entrants through successive policy revisions during the period of private sector expansion:

Pre-2007 Discretionary Regime: Coal allocation occurred through the Standing Linkage Committee (Long Term) [SLC(LT)], a body where public sector interests dominated—notably, the only consumer representative was the chairman of NTPC, India's largest public power generator (Sehgal and Tongia (2020)). These allocations, particularly through Memorandum of Understanding (MoU) based arrangements, offered generous terms including high supply guarantees and favorable pricing. Most beneficiaries were public sector plants. The discretionary nature of these allocations faced legal challenges from private generators who were kept out, leading to Supreme Court intervention that mandated a more transparent, rule-based distribution system.

2007 New Coal Distribution Policy (NCDP): The NCDP introduced a structured framework with sector-wise priorities. While improving transparency, the policy focused primarily on ensuring coal availability to the power sector rather than enhancing distribution efficiency. Under NCDP, CIL established two main supply routes: long-term fixed price contracts through Letter of Assurance (LoA) leading to Fuel Supply Agreements (FSAs), and short-term price supply through domestic spot markets or e-auctions (Ministry of Coal (2007)). Private generators, being newer entrants, predominantly fell under this regime with its stricter terms compared to the generous MoU arrangements that existing public sector plants had secured.

During the NCDP transition, CIL signed standardized FSAs with existing customers, promising 100 percent supply to the power sector and 75 percent to non-power sectors. For new projects, predominantly private, the LoA-FSA mechanism was crucial as it enabled project financing and helped secure Power Purchase Agreements (PPAs).

However, the combined commitments under FSAs and LoAs far exceeded CIL's production capabilities.

2010-2013 Coal Supply Crisis: By 2010-11, as new generators came online, most of them private, CIL struggled to meet the growing demand for coal from these generators. Recognizing this supply constraint, CIL stopped issuing new LoAs. This particularly affected private generators who were still in various stages of project development. A 2012 Presidential directive initially mandated CIL to supply 100 percent coal for 78 GW of upcoming capacity under the 12th Plan (2012-17). However, facing supply constraints, a 2013 revision reduced this to 60 GW, with 11 GW receiving tapering linkages—temporary supply arrangements that decrease as plants' captive mines become operational.[2]

Subsequent to the revised directive, the Ministry of Coal backloaded the supply guarantees and introduced differential trigger values (minimum supply guarantees)—90 percent for pre-NCDP consumers (mostly public) and 75 percent for new plants commissioned between 2009-2017 (predominantly private) until 2017[3]. The lower quota was subsequently extended till 2022. These trigger values determine penalty thresholds for supply shortfalls, effectively creating different classes of consumers based on their contracting period.

CIL linked contracted coal supply to long-term PPAs, prioritizing supply to existing plants with firm power commitments. While many private plants failed to convert their LoAs to FSAs due to strict milestone requirements, public sector plants faced fewer such hurdles. CIL continued to supply to meet demand with lower trigger values and those without firm FSAs, mostly private generators, were forced to rely more on imports.

Import-based plants: Concurrent with domestic coal allocation challenges, private generators pursuing import-based strategies faced their own policy-induced setbacks. Several large coastal power plants, such as Mundra Ultra Mega Thermal Power Project (UMTPP) and Mundra Thermal Power Station were specifically designed around imported coal to avoid dependence on India's constrained domestic supply system. The owners of most of these plants had secured mine ownership or

[2]The copy of the revised presidential directive is provided in Annexure VI of the CIL Annual Report, 2014: https://d3u7ubx0okog7j.cloudfront.net/documents/2013-14˙8w1zW4u.pdf

[3]See Ministry of Coal's notification dtd. 26 July, 2013. https://coal.nic.in/sites/default/files/2020-01/260713_0.pdf

long-term contracts at international sources, most prominently in Indonesia, allowing them to secure consistent coal supplies at prices significantly lower than international price indices.

These import-based plants enjoyed significant locational advantages. Being situated at ports eliminated India's substantial inland transportation costs, which can effectively double fuel costs for plants located 1000 km from mine sources. During the competitive bidding process in the mid-to-late 2000s, the tariffs offered by these import-based plants were competitive with those offered by domestic coal-based plants, including public sector generators, making them attractive options for state electricity boards.

However, the competitive position of import-based plants deteriorated sharply due to exogenous policy changes in coal-exporting countries. In 2014, the Indonesian government modified its coal export policies, mandating that Indonesian coal exports be priced at international benchmark rates rather than the previously negotiated long-term contract prices (Taneja (2013)). This policy shift substantially increased fuel costs for Indian import-based plants, undermining the cost competitiveness that had made them viable during the initial bidding process.

The Indonesian policy change created a supply-demand mismatch for these plants: they were locked into fixed-price Power Purchase Agreements based on their original competitive bids, but now faced substantially higher input costs. Many of these plants petitioned with CERC to secure tariff hikes, for example, both Mundra UMTPP and Mundra TPS secured compensatory tariffs of 0.52 and 0.41 INR/kWh respectively, over 20% increase of the original tariff. However, in 2017, the Supreme Court struck down the practice. Subsequent to the 2013 policy change, developers of under construction imported coal based plants including the Krishnapatnam Ultra Mega Power Plant decided to cancel the projects.

2017 SHAKTI Policy: The Scheme for Harnessing and Allocating Koyala Transparently in India (SHAKTI)introduced auction-based allocation of coal to private generators while maintaining CIL's monopoly structure (Ministry of Coal (2017)). SHAKTI closed the LoA-FSA route, requiring private sector plants without linkages but with PPAs to bid through discount offers on electricity prices. In stark contrast, the policy explicitly maintained preferential treatment for public sector plants, which continued receiving administrative allocations at notified prices without participating in any competitive process. The discounts would reduce the potential revenue

of private generators while also reducing costs. Public plants retained both secure low-cost coal access and full revenue realization from their PPAs, keeping their profits mostly unaffected. The policy's design thus perpetuated rather than eliminated the competitive disadvantages facing private generators, maintaining the barriers to private entry and investment.

Access to Coal Deposits and Captive Mining: Beyond CIL's direct coal supply, the evolution of captive mining policies delivered severe cost shocks to existing private developers while creating the most dramatic discrimination between public and private generators. In 2007, amid shortfalls in domestic coal production and spikes in imported coal prices, industry appeals to the Prime Minister (who was also the Coal Minister) led to generous allocations of coal blocks on a first-come, first-serve basis. These allocations, made at nominal prices to established and new power entities regardless of mining experience, later became controversial. The Comptroller and Auditor General (CAG) identified an opportunity-based loss of Rs. 185,591.34 crore to the exchequer, in what came to be known as "coalgate" or coal scam (Comptroller and Auditor General of India (2012)).

The 2014 Supreme Court verdict cancelling 204 out of 214 allocated coal blocks created immediate financial distress for private power developers. Many private firms had invested substantial capital in power plants specifically based on securing low-cost captive coal, with these arrangements forming the cornerstone of their project financing and securing Power Purchase Agreements. The cancellation effectively eliminated low cost coal supplies - increases in the cost of coal from CIL benchmark of 2300 INR/tonne at the time to 4000 INR/tonne for coal sourced from markets would raise variable costs by 1 INR/kWh - a 73 percent increase. These cost increases made many existing plants financially unviable while simultaneously diminishing the investment case for any new projects these firms had planned, effectively eliminating both their current profitability and incentive for future capacity expansion.

Only 14 blocks were spared from cancellation, including 12 allocated to NTPC (Pakri Barwadih) and Sasan UMPP blocks, along with one SAIL block The government's response through the Coal Mines (Special Provisions) Act, 2015, institutionalized the most explicit discrimination yet: mandating competitive auctions for private sector allocations while maintaining administrative "allotment" for public sector entities at nominal prices of INR 100/tonne for the power sector[4].

[4]Economic Times (2015).https://economictimes.indiatimes.com/industry/indl-goods/

The revised auction system, while ostensibly designed to improve transparency and coal supply, failed to address the fundamental asymmetry between public and private generators. Private generators participated in competitive auctions where they faced substantial premiums[5]—for instance, Monnet Power Company Ltd secured the Utkal C-block (with 123.86 million tonnes reserves) at INR 770 per tonne, excluding mining costs[6]. In contrast, public entities received larger blocks at nominal prices, exemplified by NTPC's access to blocks with geological reserves exceeding 7.3 billion tonnes (NIE NTPC Report, 2019).

The persistent cost disadvantage created by this system operated through a clear causal mechanism: differential coal costs → differential generation costs → differential profits → reduced private sector investment incentives. Moreover, a vast majority of private blocks were sized under 10 MT, limiting economies of scale. By December 2018, only three private sector power plants were producing from their allotted captive mines (Central Electricity Authority (2018-2023)), highlighting both the practical difficulties private generators faced in developing these competitively acquired assets and the limited success of the revised auction system in providing viable alternatives for private sector coal access.[7]..

svs/metals-mining/psus-to-pay-rs-100-150-per-ton-on-coal-besides-upfront-payment/articleshow/45979075.cms

[5]By August, 2015, 31 mines were auctioned off to the private sector, 9 of which were for power end-users

[6]Economic Times (2015).https://economictimes.indiatimes.com/industry/indl-goods/svs/metals-mining/monnet-power-company-bags-utkal-c-coal-block-for-rs-9537-22-crore/articleshow/46505049.cms

[7]These power plants include Sasan UMPP, Budge Budge TPS and Nigri TPP.

Table 1. Timeline of Coal Policy Changes and Private Sector Investment

Year	Policy Change	Impact on Private Sector
Pre-2007	Discretionary coal allocation through Standing Linkage Committee; generous MoU arrangements for incumbents	
2007	New Coal Distribution Policy (NCDP) introduces LoA-FSA mechanism	Stricter terms for new entrants; private generators face more stringent requirements
2010-11	CIL stops issuing new Letters of Assurance due to supply constraints	Private developers unable to secure coal linkages for project financing
2012-13	Presidential Directive revision reduces guaranteed supply; lower trigger values introduced (65% against 90%)	Lower coal supply guarantees
2014	Indonesian coal export policy mandates benchmark pricing; Supreme Court cancels 204 coal blocks in "Coalgate" verdict	Import-based plants face cost shocks; captive mining advantages eliminated
2015	Coal Mines Act mandates auctions for private sector vs administrative allocation for public sector	Cost advantage from private mining diminished
2017	SHAKTI policy requires private plants to bid for coal allocation; Supreme Court strikes down compensatory tariffs	Private plants lose access to fixed-price mechanisms; no recourse for cost escalations

Note: Table shows key policy changes affecting coal allocation and their differential impact on public versus private generators. The concentration of policy changes during 2013-2014 coincides with the cessation of private sector investment, considering typical 4-year construction lags.

2.2.2 The Fundamental Distortion: Absence of Efficiently Functioning Coal Markets

The convergence of domestic and international coal policy changes around 2013-2014 reflects a common underlying institutional distortion: the absence of market-based coal allocation mechanisms in India. Rather than allowing uniform pricing determined by supply and demand, India's coal system operates through administrative rationing where CIL's production decisions function independently of price signals, and access

to subsidized coal is determined by legacy arrangements and ownership status rather than productive efficiency. This fundamental distortion manifests through multiple channels—CIL's differential supply guarantees favoring public plants, discriminatory captive mining rules post-Coalgate, SHAKTI's asymmetric treatment of private generators, and the broader regulatory uncertainty that makes long-term investment planning difficult. The Indonesian policy shift mandating benchmark pricing simultaneously eliminated the import-based alternative, creating a comprehensive closure of viable business models for private thermal power investment.

This institutional structure creates systematic resource misallocation where access to subsidized inputs, rather than operational efficiency, determines competitive outcomes at both intensive margins (current profits) and extensive margins (entry and investment decisions). Given coal's 60-70% share in variable generation costs, the 2-3x price differentials between administratively allocated and market-priced coal directly translate into profit margins that favor less efficient public plants over more efficient private competitors. The mechanism operates clearly: administrative coal allocation $\rightarrow$ differential fuel costs $\rightarrow$ differential profit margins $\rightarrow$ reduced private investment incentives, explaining both the initial private investment boom (based on pre-2013 policy expectations) and the subsequent cessation as discriminatory policies became entrenched. Under a market-based allocation system, uniform coal pricing would eliminate these systematic advantages and base competitive outcomes on productive efficiency rather than regulatory position, providing the counterfactual framework for analyzing productivity misallocation in India's electricity sector.

2.3 Data

We utilize the following datasets from the 2006-18 period to construct a comprehensive monthly plant-level dataset of the electricity industry in India:

Electricity Generation: Monthly plant-level generation comes from Central Electricity Authority (CEA)'s monthly generation reports, which report operational capacity and production for all utility-scale fossil and hydroelectric, nuclear power plants in India.

Contract market data: Long-term Power Purchase Agreement (PPA) data comes from the Ministry of Power's MERIT portal. For each plant, the data reports the capacity (in MW) allocated to each buyer (typically Distribution Companies) and the corresponding tariffs (both variable and fixed).

Data on electricity trading in the short-term market comes from Central Electricity Regulatory Commission (CERC)'s *Report on Short-term Power Market in India*. These reports, published annually, report monthly volume and average price of electricity traded in the country's short-term markets, including the day-ahead market and short-term bilateral market.

Combined, these two sets of data allow us to generate plant-level panel of monthly revenue.

Wholesale electricity demand: Monthly data on state-level wholesale electricity demand comes from the Power System Operation Corporation (POSOCO) - the national grid operator that collects demand bids from state Discoms for dispatch.

Coal contracts, prices and quantities: We obtained data on the universe of active fuel supply agreements (FSAs) between power plants and Coal India Limited (CIL) from CIL. For each FSA, this data specifies the recipient power plant units, annual contracted quantity (in metric tonnes), contract signing date, contract end date and the sanctioned grade(s) of coal. Annual contracted quantity is the quantity of coal the power plant unit is eligible to receive from CIL. Actual receipts can vary depending on the domestic demand for coal and other factors such as operational constraints or rains, and plants typically receive quantities that can both exceed or remain below this contracted quantity.

We also utilize CEA's monthly coal reports that disaggregate receipts by source: Coal India Limited (CIL), Singareni Collieries Company Limited (SCCL), e-auctions, imports, and other sources. These reports also document monthly contracted quantities specified in Fuel Supply Agreements (FSAs). We validate plant receipts and contracted quantities against quantities specified in the FSA data, and find very high degree of correspondence between the two.

We construct source-specific coal prices using multiple data sources. For coal procured through FSAs (linkage-based coal), we use grade-specific notified prices from CIL and SCCL. Each plant is assigned a "minemouth" price based on the average grade (calorific value in kcal/kg) of coal it consumes. For e-auction coal, we utilize Ministry of Coal's monthly Cabinet reports that document scheme-specific premiums over notified prices and quantities. We restrict to two primary schemes through which power plants procure coal: Spot E-Auction and Special Forward E-Auction.

For imported coal prices, we use monthly price series published by the Directorate

General of Commercial Intelligence and Statistics (DGCIS). The DGCIS reports monthly quantity and value of steam coal imports (product code 27011920) by country of origin. We restrict to imports from Indonesia and South Africa as these two countries account for more than 90 percent of all imports. We validate them using a subsample of grade-specific coal import prices sourced from CoalMint.

3 Empirical evidence

The institutional distortions in coal markets created systematic differences in fuel access between public and private generators, and the evolution of coal policies hurt private sector profits, and productivity of the industry. Entering private generators in the period demonstrated superior technical efficiency. Appendix Figure 10 reveals two key patterns in heat rates. First, there is a clear vintage effect - plants commissioned before 2009, predominantly public sector, operate at heat rates exceeding 2500 kcal/kWh. Second, and more importantly, private generators consistently achieve better heat rates than public plants of the same vintage. During 2012-15, a period of substantial private entry, private plants operate at heat rates around 2200 kcal/kWh while comparable public plants require about 2400 kcal/kWh. This efficiency advantage persists across all post-2009 vintages, reflecting private plants' adoption of superior technology.

The superior technical efficiency of entering private generators becomes economically significant when translated into operational costs. Private generators consistently achieve heat rates around 2200 kcal/kWh compared to 2400-2500 kcal/kWh for public plants of comparable vintage. This 300 kcal/kWh efficiency advantage translates to 0.24-0.37 INR/kWh lower fuel costs—representing 6.7-10.4% of typical electricity tariffs. Such efficiency differences would normally provide private plants with substantial competitive advantages, enabling them to underbid public competitors in power auctions.

Figure 2 quantifies this discrimination across vintage cohorts of power plants. Pre-2009 plants, almost entirely public sector, received generous coal allocations exceeding 350 tonnes/MW. Although allocations declined for all plants commissioned after 2009, public generators maintained a significant advantage over private plants. This allocation pattern aligns with the policy evolution described above - the 2012 Presidential Directive and 2014 cancellation of coal blocks cemented these differences. Notably, substantial private entry (shown by capacity bars) occurred precisely during these

post-2009 vintages when coal access became more restrictive and discriminatory.

Figure 2. Coal allocation and receipts by ownership

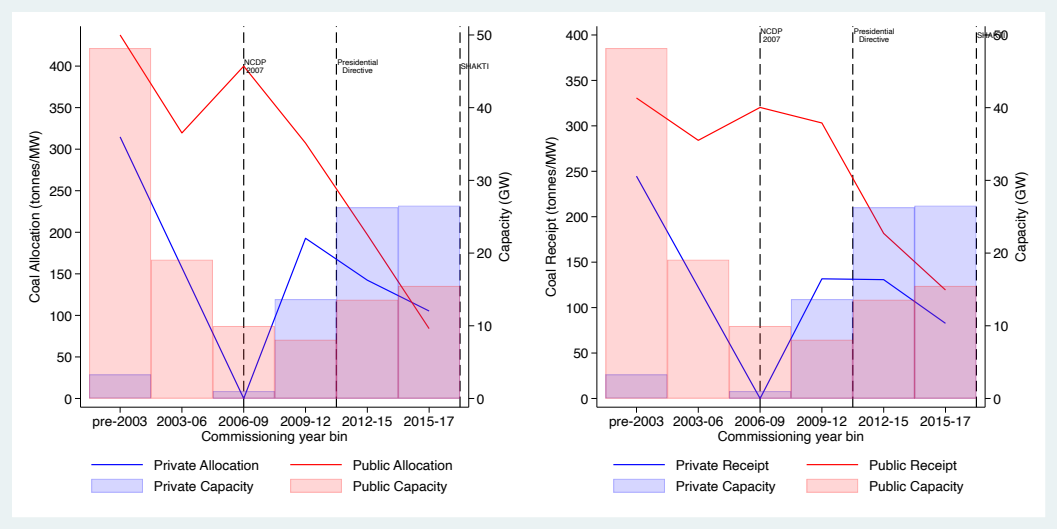


Figure shows coal allocations and receipts by plant ownership and vintage. Left panel plots CIL/SCCL coal allocation (tonnes/MW) against plant commissioning year for public (red) and private (blue) generators. Right panel shows actual coal receipts in the same format. Lines show average allocation/receipt by ownership while bars show installed capacity (GW, right axis) in each vintage bin. Vertical dashed lines mark key policy changes: New Coal Distribution Policy (2007), Presidential Directive on fuel supply agreements (2012), and SHAKTI policy (2017), explained in detail in 2.2.1.

The right panel of Figure 2 demonstrates how these allocation differences translate into actual coal receipts. Public plants commissioned post-2009 receive around 250 tonnes/MW compared to about 170 tonnes/MW for private plants of the same vintage. Importantly, differences in receipts are far larger than differences in allocation, evidence that coal supply to public power plants is prioritised even when both public and private may have allocations with CIL. Public generators source most of their coal (over 250 tonnes/MW) directly from CIL/SCCL at subsidized prices, with minimal reliance on alternatives. In contrast, private generators receive significantly less CIL/SCCL coal (about 150 tonnes/MW) and must supplement with more expensive e-auction purchases, or imports. Even after accounting for the higher quality of imported coal, private plants receive significantly less energy per MW.

These differences in receipts of subsidized coal evolved over time. Pre-2009, almost all thermal based plants were public and received nearly 90 percent of their annual coal requirement (quantity of coal needed to operate at 80% PLF) from CIL or SCCL. At the same time, many private firms had secured mining rights for their planned thermal

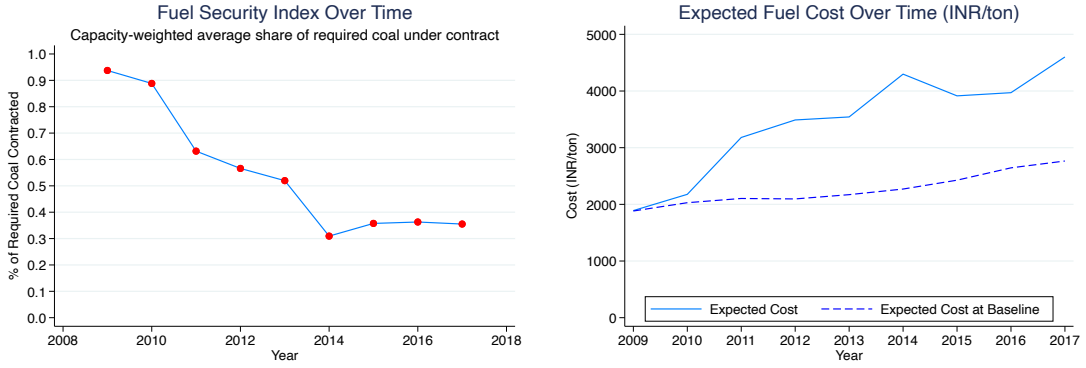
generators (what later transpired into "Coalgate") providing coal supply guarantees at competitive costs. This coal sourcing pattern formed the basis for baseline beliefs on coal access for entering private generators. Figure 3 tracks the evolution of empirically estimated beliefs about coal access. We calculate them from the gap between plants' actual coal receipts and their technical requirements for baseload operation (80% PLF). We interpret these beliefs as market participants' updating of expectations based on policy announcements, CIL's coal supply performance, and observed allocation patterns. With the evolution of coal allocation policies described in Section 2.2.1, expected domestic coal share declined dramatically from 90% to 38% between 2009-2017.

This drop in expected share of subsidized coal translated into substantial increases in expected costs, as shown in the right panel of Figure 3. By 2017, entering power plants would expect to pay over 4500 INR/ton for coal, when they'd have paid around 2900 INR/ton if the coal allocation policy stayed at pre-2009, guaranteeing 90 percent requirement be met by CIL. Given coal typically accounts for 60-70% of variable generation costs, a typical thermal plant operating at 2200 kcal/kWh heat rate would incur fuel costs of 1.72 INR/kWh at CIL's notified prices of approximately 2900 INR/tonne. A price of 4500 INR/tonne would increase these costs to 2.68 INR/kWh — a 55% increase that represents over 27% of typical electricity tariffs. Given the typical 4-year construction period for thermal plants, projects announced during 2006-2010 (based on then-current coal allocation expectations of 0.9) came online precisely during the tightening of domestic coal supply around 2012-2014. This mismatch between expectations of coal costs and policy reality explains why private capacity additions continued through 2016 despite deteriorating coal access—these plants represented the completion of projects committed before the policy landscape became clear. As a consequence, by the time of writing this paper, no successful project commitments have been made since 2014.

Later, through a regression of profits on both plant and market-level states, we show how changes in expected fuel costs translate into lower profits for existing private generators.

Figure 3. Evolution of plant expected coal costs

(a) **A.** (b) **B.**



The left panel shows the evolution of empirically estimated power plant beliefs about the share of their coal requirement (when operating at 80% capacity factor) that would be supplied by Coal India Ltd at notified prices. These beliefs are estimated from observed gaps between plants' actual coal receipts and requirements. The right panel translates these beliefs into expected weighted-average coal costs in INR/tonne. The solid blue line shows expected coal costs calculated by weighting CIL's notified prices by the belief share in the left panel, with the remaining share priced at prevailing imported coal rates (adjusted for calorific value differences). The dashed blue line shows counterfactual costs if plants received 90% of their requirement from CIL at notified prices, representing pre-2009 baseline expectations.

4 Model and Estimation

4.1 Model

We develop an infinite-horizon dynamic oligopoly model of investment in electricity generation capacity. Each period is four years because it takes on average four years to build new coal capacity; using one-year periods would require tracking investments made in past years that are yet to mature, making computation more difficult. At the beginning of each period t, each firm j updates its beliefs $\mu_t, \mu_{t+1}, \mu_{t+2}, \cdots$ about coal policy; this belief path means firms believe share μ_s of period-s investment will be supplied by domestic coal, when it comes online in period $s+1$, for each period $s \geq t$. Note that coal policy can vary over time. Then, observing the industry state s_t (also observeable to the researcher) and privately known i.i.d. investment cost shocks ϵ_{jt}, firms make costly investment decisions simultaneously. Investment and coal policy affect the evolution of industry states into next period. Having made investment decisions, firms take the current fleet of generators as given, and earn profit by generating electricity under dispatch by system operators.

We do not model entry and exit of firms directly. There was hardly exit in our data. While the number of private firms increased substantially, we took the set of active firms in the last year in the data (2017) to be the set of all firms present from the beginning. This modeling decision allows us to focus on capacity expansion as a margin of industry growth, and circumvents the difficult question of how many potential entrants to allow for per period, who would draw entry cost shocks.

Solving this dynamic model requires confronting the curse of dimensionality. In a conventional Markov-Perfect Equilibrium, the industry state consists of every firm's states $s_t = (s_{1t}, s_{2t}, \cdots s_{Jt})$; in our context, the s_{jt} includes capacity and domestic coal access. With around $J = 10$ players per market in our case, the state space becomes prohibitively large.

To reduce the dimension of the state space, we use Approximate Belief Oligopoly Equilibrium (ABOE), developed in Gowrisankaran et al. (2025). Instead of requiring each firm to track everyone's states, we assume they track (1) their own states s_{jt}, and (2) a set of aggregate market states Ω_t. We choose states to summarize important determinants of profits. All firms understand that their own action can influence future aggregate market states, a distinctive feature of ABOE relative to other equilibrium concepts that reduce the state space. ABOE also imposes a consistency requirement on how the reduced set of states (Ω_t, s_{jt}) evolve, conditional on firms' actions. Specifically, they approximate, with simple functional forms, the exact state transitions implied by equilibrium strategies.

For our setting, we choose two aggregate states: (1) coal share, defined as total coal capacity, divided by forecasted electricity demand in 5 years; and (2) the ratio of domestic (subsidized) coal price to the international price. We choose these states because they are plausibly important determinants of per-period profit. Starting with coal share, recall that firms earn most revenue on long-term PPA contracts, which are determined in auctions run by regulators. When coal capacity is scarce, coal share is low, and regulators are willing to pay more to grow the electricity sector. So lower coal share would be associated with higher PPA prices, all else constant. Importantly, this aggregate variable is endogenous, for more investment this period would lead to higher coal share next period, all else constant.

The second aggregate state is exogenous, the ratio of domestic to international coal prices. This ratio captures the cost advantage of having greater domestic coal access. A firm's fuel cost is a weighted average of the domestic coal price and the interna-

tional coal price. The greater the domestic coal access, the greater the weight on domestic coal price, which is usually lower, and so the marginal cost will be lower too. While marginal cost affects profit directly, for any given quantity, it also affects profit indirectly, through quantity. A firm with lower marginal cost is more likely to be called upon by the system operator to produce.

We now summarize our specification of ABOE. Starting with firm-specific states s_j, (1) capacity; (2) domestic coal access, we let

Definition 1 (ABOE). *1. Aggregate market state Ω consists of (1) Ω^1, coal share and (2) Ω^2, ratio of domestic to international coal price.*

2. A strategy $\tilde{\sigma}_j$ for each player j maps states (Ω, s_j) and investment cost shock ϵ_j to an investment level ΔK.

3. Perceived transition probabilities for aggregate states Ω, contingent on player actions, should approximate transitions induced by the strategies, up to an AR(1) regression.

4. Strategies $\tilde{\sigma}_j$ maximize expected discounted payoffs, relative to perceived transition probabilities.

In principle, strategies can be indexed by players, and can depend on persistent, player-specific features. For example, we observe some players having lower heat rate, and some locating closer to mines than others. Such players would have lower cost and their investment behavior may differ systematically from others, even when facing the same aggregate states, capacity, and domestic coal access. For now, to speed up computation, we require that all firms use the same strategy function.

4.2 Per-period profit

We first explain how firms earn per-period profits. A firm owns a set of generators. Most electricity is sold under long-term PPA contracts, awarded through auctions run by regulators, where firms bid to supply some quantity at a given price. From day to day, firms compete by making their generators available to a system operator, who allocates production to meet demand, taking into account PPA contracts and system constraints. A small fraction of capacity sells electricity in a spot market, rather than under PPAs. On the cost side, firms procure coal, either domestically or internationally, required for generating electricity. They also pay for fixed maintenance costs to

keep generators in good working condition.

Rather than model per-period profit structurally, we estimate it directly as a function of model states. It is quite standard in industrial organization to first estimate demand and marginal cost functions under price-setting assumptions, and then derive equilibrium profit. We do not take this approach because micro-founding per-period profit is complex. It requires modeling PPA auctions, as well as how the system operator allocates production, both challenging problems. [8] Additionally, we are mainly interested in per-period profit because it constitutes an ingredient in the dynamic model of investment. Knowing how per-period profit varies with states is sufficient for solving the dynamic model, even without modeling the underlying mechanisms that determine per-period profit. Thus, we deviate from the standard approach and directly estimate the per-period profit as a function of states that determine the environment for the PPA auction and day-to-day production. [9] We assume that in counterfactuals, the per-period profit function remains fixed, much as supply and demand functions are assumed fixed across counterfactuals in standard IO approaches.

Detailed PPA, fuel contract, and generation data make this approach feasible. Observing key determinants of profit makes it plausible to compute profit. For each generating unit and year-month, we compute profit using

$$\text{PPA Revenue} + \text{Spot-market Revenue} - \text{Fuel Cost} - \text{Fixed Cost}$$

We can compute revenue because we observe PPA prices, spot-market prices, and how much the generators sell under each avenue. To compute fuel costs, we observe how much coal consumption is satisfied by domestic coal, and how much by international coal; then we combine coal usage with coal prices. Finally, keeping generators operational incurs fixed costs, e.g. maintenance, which we proxy with regulators' estimates.

Rich data also allow us to construct predictors of profit: two aggregate states and two firm-specific states. Generator capacity is observed directly. To construct domestic coal access, we first use the coal supply contracts to get the quantity of coal supplied domestically; then, we use heat rate and capacity to infer how much coal is required for the generator to operate at 80 percent Plant Load Factor. Dividing contracted

[8] Ryan (2012) develops a bidding model to study PPA auctions.

[9] This treatment parallels Gowrisankaran et al. (2025), who estimate profit surfaces and use them as inputs to a dynamic model of generator decisions.

domestic coal by the coal required to operate gives us domestic coal access. The two aggregate states require less granular data. To construct coal share, we take the total industry capacity of coal generators and divide by demand forecasts published by regulators (Electric Power Surveys). Lastly, to construct coal price ratios, we take domestic and international coal prices that we collected above to compute profit.

To match firm-period-level investment decisions in the dynamic decisions, we aggregate generator-level data. We do so because plausibly firms, rather than generators, make investment decisions. If a firm has limited management ability, it may be unable to run four generators as efficiently as two, exhibiting decreasing returns to scale. So we estimate profit using firm-monthly level data. On the other hand, the dynamic model operates at the firm-period level, so we take monthly profit and scale up to match the length of one period (48 months). This allows us to use more observations when estimating the profit function and get more precise estimates.

Having constructed profits and states, we estimate profit as a function of states, using a random forest. The main benefit is that this allows for flexible functional forms. Profit can exhibit decreasing returns to capacity; domestic coal access may impact profit through its interaction with capacity, and perhaps even with the ratio of domestic to imported coal prices. Since it is difficult to specify these relationships correctly, we take a data-driven approach. The downside of this approach is that estimates are harder to interpret than those from a regression. To understand results, we will quantify the importance of predictors, and plot the estimated profit function over states commonly seen in the data.

4.3 Dynamic model

Equipped with an estimate of the per-period profit function $\pi()$, we turn to the dynamic model. In period t, each firm j observes the aggregate state Ω_t and its own firm-specific states s_{jt}. Firms draw i.i.d. private investment cost shocks $\epsilon_{jt} \sim \sigma N(0,1)$, and choose an investment level ΔK. Investment becomes operational next period.

We define the choice set to be 8 evenly spaced grid levels ranging from 0 to 1.2 times the largest observed capacity. This wide coverage helps encompass plausible levels of investment, although it comes at a cost of making the grid coarser than observed

investment levels. Investing ΔK incurs a cost of

$$\Phi^{invest}(\Delta K) = \kappa^F 1[\Delta K > 0] + (\kappa^1 + \epsilon_{jt}^{invest})\Delta K$$

The unobserved shock to marginal investment cost can help explain why firms in identical states choose different investment levels. An alternative investment cost specification would include additive, action-specific logit shocks that are independent across actions. The multiplicative form we use has two advantages (Gowrisankaran and Schmidt-Dengler (2025)): it generates more plausible substitution patterns between investment decisions, and it tends to be more well-behaved as we refine the grid.

Now we specify the Bellman equation for firm j in period t. We observe three periods of data, from 2006 to 2017, over which coal policy evolves, and we assume in period 3, firms believe the coal policy remains stationary afterwards. The perceived path of coal policy is hence $\mu_1, \mu_2, \mu_3, \mu_3, \mu_3, \cdots$. We will estimate actual coal policy from the data. As a benchmark, we assume firms know throughout the entire path of actual coal policy, ruling out surprising coal policy changes. [10]

When aggregate market states are Ω_t and firm specific states (1) capacity K_{jt} and (2) domestic coal access shr_{jt}, investment strategy $\tilde{\sigma}_j(\Omega_t, K_{jt}, shr_{jt}, \epsilon_{jt}; t)$ should solve the maximization problem

$$\begin{aligned}\tilde{V}_j(\Omega_t, K_{jt}, shr_{jt}; t; \epsilon_{jt}^{invest}) = \max_{\Delta K} \Pi_j(\Omega_t, K_{jt}, shr_{jt}) - \underbrace{(\kappa^F 1_{\Delta K>0} + \kappa^1 \Delta K)}_{\text{investment cost (deterministic)}} - \epsilon_{jt}\Delta K + \\ \beta E[V_j(\Omega_{t+1}, K_{j,t+1}, shr_{j,t+1}; t^+) \mid \Omega_t, K_{jt}, shr_{jt}, \Delta K] \\ \equiv \max_{\Delta K} \underbrace{\tilde{V}_j^{\Delta K}(\Omega_t, K_{jt}, shr_{jt}; t; \epsilon_{jt}^{invest})}_{\text{Choice-specific value}} - \epsilon_{jt}\Delta K\end{aligned}$$

where

$$t^+ = \begin{cases} t+1, \text{ if } t < 3 \\ t, \text{ if } t \geq 3 \end{cases}$$

and $V_j(\Omega_t, K_{jt}, shr_{jt}; t)$ denotes the ex-ante value, obtained by integrating over in-

[10] An opposing alternative would be to assume myopic beliefs, where in each period t, a firm believes future coal policy would remain fixed at μ_t for good.

vestment cost shocks ϵ_{jt}^{invest}. Formally,

$$V_j(\Omega_t, K_{jt}, shr_{jt}; t) = \int \tilde{V}_j(\Omega_t, K_{jt}, shr_{jt}; t; \epsilon_{jt}^{invest}) dF(\epsilon_{jt}^{invest})$$

for each period t being considered. Similarly, integrating investment strategies $\tilde{\sigma}_j$ over investment cost shock yields conditional choice probabilities σ_j.

The j subscript and t argument require clarification. In principle, each firm j can have its own value function, based on firm-specific, time-invariant features, including, for example, firm-specific average heat rate. In our current implementation, we do not include such features, so V_j is the same function for all j.

Second, the t argument in the value function tracks which coal policy parameters remain relevant. In period 1, the entire path $\mu_1, \mu_2, \mu_3, \mu_3, \cdots$ is relevant; in the right-hand side, the continuation value captures payoffs starting from period 2, from which only $\mu_2, \mu_3, \mu_3, \cdots$ are relevant. From period 3 onward, we no longer advance the period argument on the right-hand side because we assume beliefs about future coal policy become stationary afterwards; the future path of coal policy remains fixed at μ_3. This suggests how we would solve the bellman equations. Starting from $t = 3$, we solve the Bellman by fixed point iteration; and then we can solve for the period-2 Bellman by backward induction from the period-3 Bellman. And similarly for the period-1 Bellman.

Next, we explain how firm-specific states evolve across periods. Capacity increases by the investment level ΔK. The other firm-specific state, domestic coal access, evolves based on investment ΔK and coal policy beliefs μ_t. It is easier to discuss the transition process in terms of the total capacity that has domestic coal access. This capacity next period consists of three components. First, existing coal contracts survive, except with a cancellation probability that is zero in all periods except period two; this probability is required to match observed period-2 contract cancellations for firms implicated in the Coalgate scandal. Second, existing capacity that does not have domestic coal access may gain such access. Third, new investment may acquire some degree of domestic coal access. We let μ_t denote all three components, with the

understanding that the third component is of greatest interest. Formally,

$$shr_{j,t+1}(K_{jt}+\Delta K) = \underbrace{shr_{jt}K_{jt}(1-\text{Cancelled}_t)}_{(1)} + \underbrace{(1-shr_{jt})K_{jt}\text{PolicyBacklog}}_{(2)} + \underbrace{(\Delta K)\text{PolicyNew}_t}_{(3)}$$

Unlike firm-specific states, aggregate states Ω_t evolve stochastically. The state Ω^2, ratio of domestic to imported coal price, evolves exogenously according to an AR(1) regression. The other aggregate state, Ω^1, coal share, evolves endogenously. In ABOE, players approximate the true transition, induced by players' investment strategies, by a regression. Using

$$\Omega^1_{t+1} = a_0 + a_1\Omega^1_t + a_2\Omega^2_t + a_3\Omega^1_t\Omega^2_t + e^1_t$$

a firm forecasts Ω^1_{t+1}, conditional on its investing the mean investment level $E(\Delta K)$ implied by its choice probabilities. The firm then obtains choice-specific forecasts of Ω^1_{t+1} by adjusting this forecast by the difference between ΔK and $E(\Delta K)$, normalized by forecasted demand. Each firm recognizes that by increasing ΔK, it would raise Ω^1_{t+1} for all firms.

Given investment cost parameters and the transition equation for domestic coal access, we can solve for ABOE using a nested fixed-point algorithm. Our goal is to solve for ex-ante value functions $V_j(\cdot)$, conditional choice probabilities σ_j, as well as the state transition coefficients $a \equiv (a_0, a_1, a_2, a_3)$.

Algorithm 1: Nested fixed-point algorithm for computing ABOE

Data: Investment-cost parameters, the per-period profit function, and the law of motion for domestic coal access

Result: Ex-ante value functions V_j, choice probabilities σ_j, and aggregate-state transition coefficients a

Initialize V_j, σ_j, and a;

while the a (aggregate-state transition coefficients) have not converged **do**

  Given a, solve the Bellman equation for V_j and σ_j for every firm j;

  Using σ_j, simulate firms' choices and the resulting future states;

  Re-estimate the aggregate-state transition regression on the simulated states to update a;

end

4.4 Estimating the dynamic model

We need to estimate coal policy, i.e., the law of motion for domestic coal access, and more importantly, investment cost parameters. To estimate the former, we use observed domestic coal access and capacities, without solving the dynamic model. To estimate Cancelled_t, we measure the change in domestic coal access before and after Coalgate cancellations, which occur in period 2; we set this to zero in other periods. We estimate the other coal policy coefficients by least squares.

We estimate investment cost parameters $(\kappa^F, \kappa^1, \sigma)$ by nested fixed-point GMM. We search for parameters to match model-implied moments with their data analogs. To identify the fixed cost of investment, we look at the probability of investing. To identify κ^1, we look at mean investment. To identify σ, we look at variance in investment. We also look at the interaction between capacity and an indicator for investing, which captures how investment varies with firm-specific states.

4.5 Electricity supply curve and welfare

The dynamic model delivers private capacity and domestic coal access by market and period. To translate these into output, generation costs, and shortages, we build a monthly supply curve for each regional market and clear it against observed demand. Let $r \in \{N, W, S, E\}$ index markets and m index months within period t. Each market contains coal generating units $i \in \mathcal{I}_r$ with capacity K_{im} and constant marginal cost within each month

$$c_{im} = h_i \left[s_i \frac{P^D_{im} + T_i}{\gamma^D} + (1 - s_i) \frac{P^I_m + T_i}{\gamma^I} \right],$$

where h_i is the heat rate, s_i the share of coal bought at the notified domestic price P^D_{im}, P^I_m the import price, T_i the per-tonne transportation cost from mine or port, and γ^D, γ^I the calorific values of domestic and imported coal. Coal and gas serve net demand $R_{rm} = D_{rm} - \bar{H}_{rm} - \bar{N}_{rm}$, demand less hydro and nuclear generation, which

we take as given rather than model explicitly (see below). The market solves

$$\begin{aligned} \min_{\{q_{im}\},g_{rm}} \quad & \sum_{i\in\mathcal{I}_r} c_{im}q_{im} + c_m^G g_{rm} \\ \text{s.t.} \quad & \sum_{i\in\mathcal{I}_r} q_{im} + g_{rm} = \min\Big\{R_{rm},\ \sum_{i\in\mathcal{I}_r} \bar{\alpha}K_{im} + \bar{G}_{rm}\Big\}, \\ & 0 \le q_{im} \le \bar{\alpha}K_{im}, \qquad 0 \le g_{rm} \le \bar{G}_{rm}, \end{aligned}$$

where q_{im} is coal generation and g_{rm} gas generation. Generation meets net demand when available capacity allows, and otherwise uses all available capacity. The solution fills coal units in order of marginal cost, then gas, and unmet demand is $u_{rm} = R_{rm} - \sum_i q_{im} - g_{rm}$. The parameter $\bar{\alpha}$ is the share of installed coal capacity available to generate. We set it to 0.8, the average available capacity net of auxiliary consumption and the outages we observe in the data.

The cost of the last unit used, plus a period-specific shift σ_t, gives the model's short-term market price. The shift is the mean gap between the PPA variable charge that plants are paid and their fuel cost (INR 0.47/kWh in 2010–2013 and 0.75/kWh in 2014–2017); it moves the price level but not the order of the supply curve. This is the price at which electricity trades in the short-term market, on power exchanges and through short-term bilateral contracts. It is only one part of what wholesale buyers pay: about 90 percent of electricity is sold under long-term PPAs, with their own fixed and variable charges (Central Electricity Regulatory Commission (2015)). The short-term price should therefore not be read as the average wholesale price of electricity in India.

We clear the supply curve against peak demand, which includes the observed deficit, because shortages in India were concentrated in the evening peak. Other energy sources enter as follows. Nuclear plants run at base load, so nuclear output $\bar{N}_{rm}$ is held at its observed level. We do not model hydro generation explicitly. It is seasonal, limited by the water stored in reservoirs, and typically load-following: reservoirs are released to meet demand, so hydro output varies over the day, with the highest capacity utilization in peak hours (Carbon Tracker Initiative (2026)). Observed monthly output, an average over all hours, therefore understates the hydro available at the peak. We set $\bar{H}_{rm}$ to 1.5 times observed output, the ratio of peak to off-peak hydro generation in the intraday profile, capped at installed hydro capacity. Gas comes last, up to its observed output $\bar{G}_{rm}$: gas plants were fuel-constrained after 2011, and we

lack data on their costs. Renewables and the North-Eastern market are excluded, as both were small over this period. Markets clear separately, except that central-sector coal plants in the East, which are contracted to states across India, enter each market's supply curve in proportion to their PPA allocations to that market.

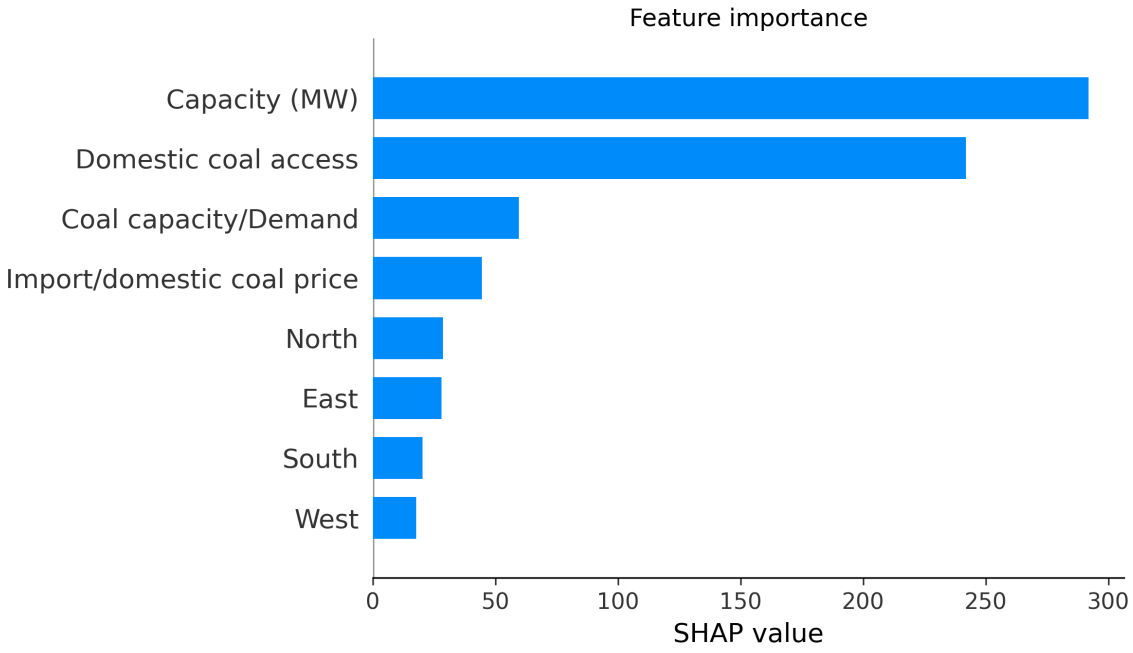
In the counterfactuals, public plants and import-based private plants keep their observed capacities and costs, unit by unit. Private non-import capacity is replaced by one representative unit per market, with capacity and domestic coal access s_{rt} from the dynamic model, the average heat rate of private plants in the market, and the average transportation cost of private non-import plants in the market and period. Import-based private plants buy no domestic coal under the factual policy ($s_i = 0$) and receive s_{rt} under the counterfactual policies; because they burn imported coal at their port, they incur no inland transportation cost. The dynamic model gives investment for each four-year period but not its timing within the period. We therefore divide each period's new capacity into four equal parts and bring one part online at the start of each year of the period. We apply the same rule to new public and import-based capacity, so that all capacity is treated alike. We compare every counterfactual with the factual simulation under the same assumptions, so that level errors in the supply curve largely cancel.

The counterfactual coal policies are subsidies: the government pays private generators the gap between the market and the notified price on their subsidized share of coal. Let C be the cost of coal and gas at the prices generators pay and U unmet energy. The welfare change from the factual policy (F) to a counterfactual (CF) is

$$\Delta\text{Welfare} = \underbrace{\left(C^F - C^{CF}\right) - (S + X - E)}_{\text{productive efficiency, net of subsidy}} + \underbrace{W\left(U^F - U^{CF}\right)}_{\text{lower shortages}},$$

where S is the subsidy on private output under the factual policy, X the subsidy on additional private output, and E the value of domestic coal no longer burnt by public plants, which is sold at the market price because total domestic coal output is fixed. Netting the subsidy removes the mechanical saving from buying coal at the notified price. What remains is the saving from shifting production to generators with lower heat rates and transportation costs, and from replacing gas. We set W to INR 5 per kWh, close to the average residential tariff in 2014–15 and below commercial tariffs (Central Electricity Authority (2015)). We read the resulting gains from lower shortages as conservative, because residential tariffs are subsidized and lie below the

Figure 4. Importance of predictors for per-period profit



cost of supply, and a tariff is a lower bound on what a rationed consumer would pay for the lost electricity. Moreover, many of the consumers at the margin of shortages are industrial and commercial firms, which suffer significant economic losses from outages and so value reliable supply well above the tariff (Allcott et al. (2016)).

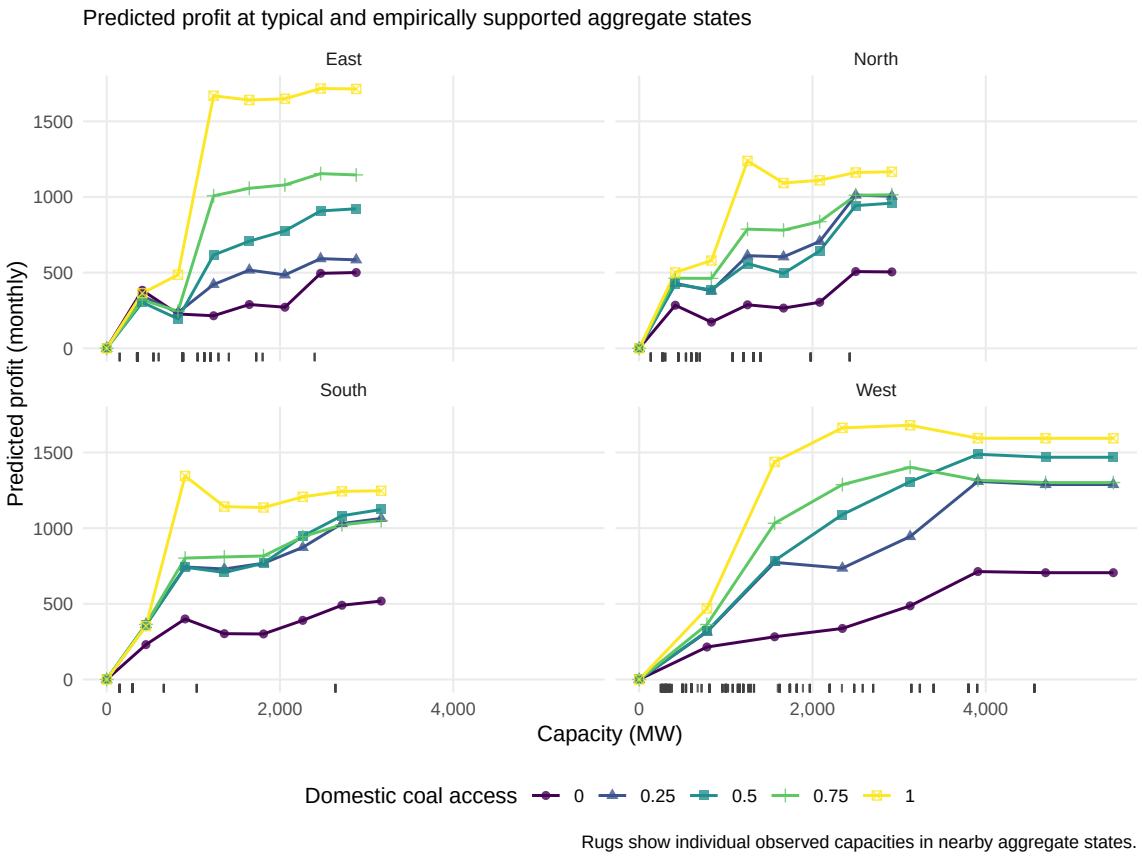
5 Results

5.1 Results for per-period profit surface

We begin by illustrating our estimates for the per-period profit surface, a key input into the dynamic model. Our estimates achieve an out-of-sample R^2 of 0.83. We illustrate our results in two ways. First, the bar plot shows the relative importance of predictors, quantified by SHAP values. They measure the average absolute contribution of each predictor to predicted profit. The two player-specific states are most influential, followed by the aggregate states and market indicators.

Second, we plot predicted profit in each market. Profit being predicted from four states, to visualize it we hold two aggregate states at representative values in each market; then, we trace out predicted profit over capacity, with one line for every level of domestic coal access.

Figure 5. Predicted per-period profit at representative aggregate states



Some characteristics of the fitted profit function agree with economic intuition. Within market, predicted profit tends to rise with capacity and with domestic coal access. This makes sense because additional generation capacity is procured only if it is expected to turn a profit, and greater domestic coal access tends to reduce the cost of fuel. Second, at least when capacity is low, the higher the domestic coal access, the faster predicted profit rises with capacity. This suggests complementarity between capacity and domestic coal access, which makes sense because domestic coal access enters into profit through fuel cost, and the greater the capacity, the more coal is required, changing domestic coal access would have greater impact on total fuel cost.

Next, we discuss less intuitive characteristics of the fitted profit surface. First, predicted profit tends to flatten out in capacity at relatively low capacity levels. Second, predicted profit exhibits a spike around $1000MW$ in some markets. These patterns arise in part because the random forest is very flexible, and we impose no monotonicity constraint.

One reason that profit flattens out in capacity is that profit at moderate capacities around 1000 MW is remarkably high. Such profits were realized early on, when spot-market prices peaked, by players who sold largely to the spot market. Over the course of our sample, two changes occurred: these players invested in additional capacity, and spot market prices declined to less than half their peak value (from around 8 INR/KWh to 3 INR/KWh). Thus, a player's profit could be lower when its capacity was higher.

A second explanation for the shape of the profit surface relates to how random forest predicts profit at state vectors sparsely covered by the data. The rug beneath each figure suggests that we observe a wide range of capacities, but higher capacity often coincides with moderate or low domestic coal access. For full domestic coal access, we mostly observe capacity up to half the maximal value, making it challenging to predict its trajectory if capacity should continue to rise.

5.2 Results for parameters in the dynamic model

We first present estimates of coal policy, which govern the transition of domestic coal access. We estimate the transitions separately for private and public firms. As Table 2 shows, the private firms consistently receive lower domestic coal access on new capacity. The private–public difference is modest in periods 1 and 3, however, and considerably larger in period 2. The evidence is consistent with public firms

getting preferential allocation of subsidized coal, although the disparity is not equally pronounced across periods.

Another pattern is that for both private and public firms, coal policy is most generous in period 2, and much more stringent in the other two periods. This pattern is consistent with evidence that regulators observed slow contract signing in period 1, and subsequently urged coal suppliers to sign domestic coal supply contracts with new generators.

Table 2. Coal policy estimates

	Private	Public
New capacity, period 1	0.37	0.43
	(0.11)	(0.06)
New capacity, period 2	0.54	0.74
	(0.05)	(0.06)
New capacity, period 3	0.30	0.37
	(0.14)	(0.06)
Previously uncontracted capacity	0.25	0.09
	(0.07)	(0.04)
Period-2 cancellation rate	0.32	–
	[Fixed]	

Notes: Standard errors are in parentheses. The new-capacity coefficients give the share of capacity added in each period that obtains domestic coal access. The backlog coefficient gives the share of previously uncontracted capacity that obtains access. The private specification fixes the period-2 cancellation rate at 0.32; the public specification does not include a cancellation term.

Table 3 presents estimates for investment cost parameters. Unrestricted estimates for all three parameters are imprecise. So we provide a second set of estimates that calibrate $\kappa^1 = 150$ using Elliott (2025), who studies capacity investment in the Western Australian electricity market. He uses investment costs for coal generators of around AUD 3.448 million per MW in 2010. This translates into around INR 150 million per MW, slightly lower than our estimate.

Table 3. Investment cost estimates

	Unrestricted	Calibrated $\kappa^1 = 150$
κ^F Fixed investment cost	-180,227	-96,862
	(325,180)	(17,998)
κ^1 Linear investment cost	236.58	150.00
	(329.10)	[Calibrated]
σ Standard deviation of shock	157.43	90.22
	(262.16)	(16.57)
GMM criterion	0.836	1.039

Notes: Standard errors are in parentheses. The calibrated-column standard errors condition on $\kappa^1 = 150$. Both columns use the equal-weight private coal-contract transition law with new-capacity rates $(0.37, 0.54, 0.30)$, backlog rate 0.25, and period-2 cancellation rate 0.32. Profits and costs are measured in millions of Indian rupees, and capacity is measured in MW. Thus κ^F is in millions of rupees, while κ^1 and σ are in millions of rupees per MW. The reported standard errors use a 0.5 percent finite-difference step and do not account for the error incurred in estimating the per-period profit function.

Both sets of estimates find large, negative fixed costs of investment. One possible interpretation is that it absorbs investment incentives not captured by the model.

To understand model fit, we compare in Table 4 model-implied moments with their data analogs. The model fits the data well on the probability of investing, the mean investment, and the variance of investment. However, it underpredicts by around 15% the interaction between investing and capacity. The model underpredicts the concentration of investment among firms with greater existing capacity. The shape of our per-period profit function may shed light on this difficulty: since decreasing returns to capacity are strong, at positive capacities, it becomes unappealing to invest more.

Table 4. Fit of the preferred model: estimation moments

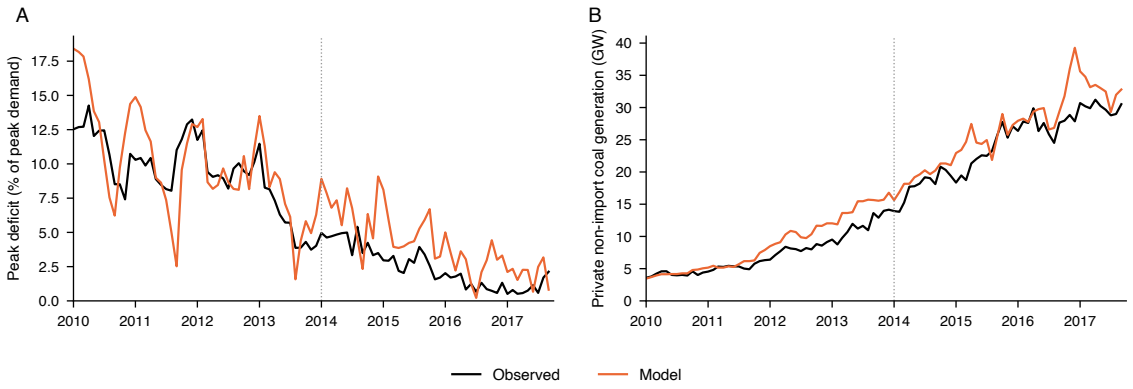
Moment	Data	Model	Model − data
$\Pr(\Delta K > 0)$	0.7549	0.7599	0.0050
$\mathrm{E}[\Delta K]$ (MW)	510.45	514.82	4.37
$\mathrm{Var}(\Delta K)$ (10^6 MW^2)	0.3891	0.4008	0.0117
$\mathrm{E}[1\{\Delta K > 0\}K]$ (MW)	407.01	344.77	-62.24

Notes: Model moments use the preferred specification with $\kappa^1 = 150$, $\widehat{\kappa}^F = -96,862$, and $\widehat{\sigma} = 90.22$, under the corrected equal-weight private coal-contract transition law.

We also validate the supply curve of Section 4.5, which translates investment into output, costs, and shortages. Holding capacity and costs at their observed values, we clear each market's supply curve every month from 2010 to 2017 and compare

outcomes with the data. The supply curve reproduces plant-level output well: the R^2 of predicted on observed monthly generation across plants is 0.77 in 2010–2013 and 0.68 in 2014–2017. Figure 6 shows two aggregate outcomes central to our counterfactuals. Panel A shows that the model tracks the decline in peak shortages, with a mean deficit of 10.0 percent of peak demand in 2010–2013 (9.3 percent observed) and 4.2 percent in 2014–2017 (2.4 percent observed). Panel B shows that it tracks the rise of private non-import generation, the margin our counterfactual policies move, with a monthly correlation of 0.98; it slightly overpredicts its level, at 8.8 GW against 7.3 GW observed in 2010–2013 and 26.5 GW against 24.1 GW in 2014–2017.

Figure 6. Fit of the supply curve: peak shortages and private generation

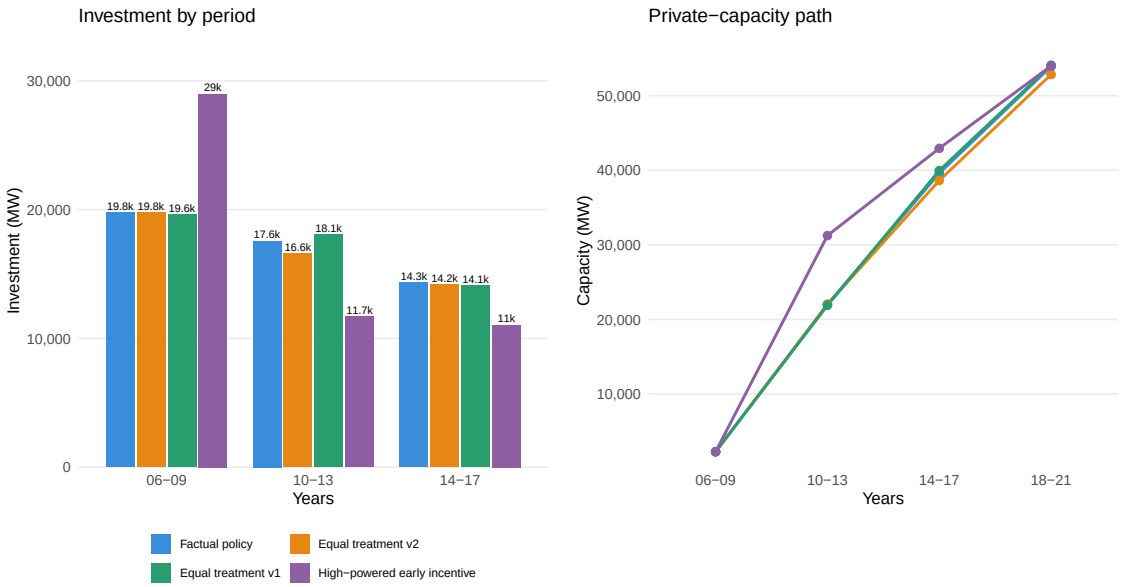


Notes: Monthly, all four markets combined. Panel A: unmet peak demand as a percentage of peak demand; observed values are from CEA. Panel B: average generation of private coal plants that do not run on imported coal, over plant-months with observed generation. The dotted line marks the boundary between periods 2 and 3.

5.3 Results for counterfactuals

Equipped with investment cost estimates, we conduct two sets of coal policy counterfactuals. The first set of counterfactuals seeks to remove policy-induced cost disparities between public and private generators. The second set recognizes that supply shortage was a serious concern in India in our sample period, and seeks to accelerate capacity expansion. For each counterfactual, we first report its effect on investment and capacity. We then feed the simulated capacity and domestic coal access into the supply curve of Section 4.5, and compare outcomes with the factual simulation over 2010–2017. Figure 7 shows investment and capacity, Figures 8 and 9 show annual shortages and average wholesale electricity costs relative to the factual policy, and Table 5 decomposes the welfare impacts.

Figure 7. Counterfactual investment and capacity



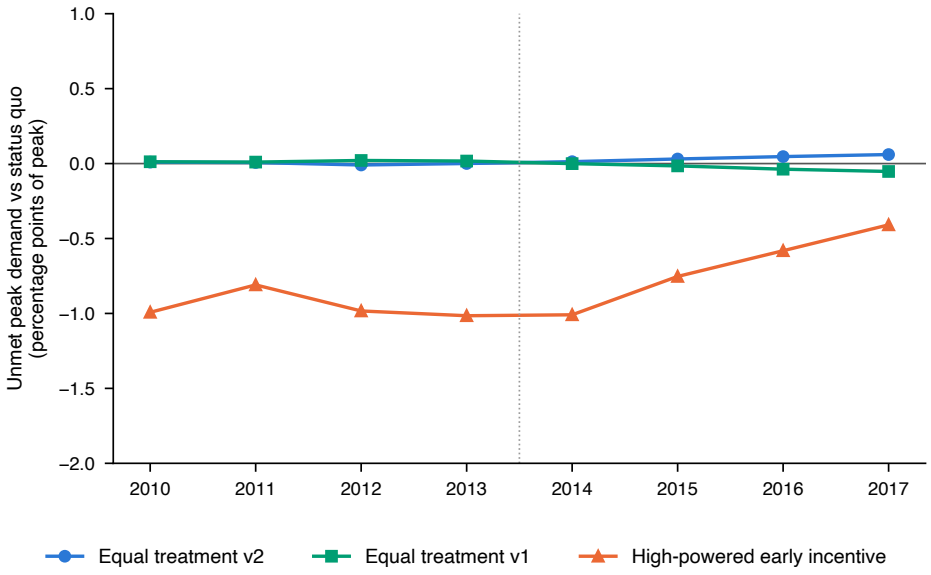
Notes: Terminal private capacity in 2018–21 is 53,921 MW under factual policy, 54,088 MW under Equal treatment v1, 52,834 MW under Equal treatment v2, and 53,984 MW under the high–powered early incentive. Differences from factual are +168, −1,087, and +63 MW, respectively.

We implement the first set of counterfactuals in two ways. First, we estimate how domestic coal access evolves for public generators in our sample, and apply the estimates to private generators ("v1" in Figure 7). Under such domestic coal access, reported in Table 2, we find that by the end of the sample period, total private investment would be 168 MW higher relative to the factual scenario. Breaking down by period, we see this counterfactual produces around 500 MW more investment in period 2, and less investment in periods 1 and 3. Notably, raising domestic coal access on new investment for every period need not increase investment in all periods; since period-2 policy is better than period-1, period 1 players have an incentive to postpone investment until period 2. This helps explain the seemingly paradoxical result that period-1 investment declines slightly, despite the policy becoming better.

Because v1 barely changes total private capacity, its welfare effects come almost entirely from reallocating production across existing generators. Unmet peak demand stays within 0.1 percentage points of the factual policy in every year (Figure 8). Raising private generators' domestic coal access lowers their marginal cost, moving them down the supply curve, so production shifts from public plants to private ones. Much of this shift goes to import-based private plants. These plants burn imported coal at their port and so incur no inland transportation cost, whereas the public

generation they replace pays rail transportation costs of around INR 0.6/kWh. Most of the measured fall in generation cost (Figure 9) reflects coal bought at the subsidized price, a transfer from the government. Net of the subsidy, productive efficiency rises by INR 10.3 billion a year, mostly from lower transportation costs (INR 9.0 billion) and to a lesser extent from lower heat rates (INR 1.4 billion). The net welfare gain is INR 10.6 billion a year.

Figure 8. Unmet peak demand relative to the factual policy



Notes: Unmet peak demand in each counterfactual minus the factual simulation, in percentage points of peak demand, all four markets combined. Negative values mean fewer shortages than under the factual policy. 2017 covers January to September.

Second, we remove the waiting incentive by applying a uniform coal policy across all periods. Specifically, we compute the capacity-weighted average of public players' access to domestic coal in our sample period (0.72) and apply this to new private investment in each period ("v2" in Figure 7). We also raise initial domestic coal access (in period 1) to at least 0.72, for otherwise private players with poor initial coal access would not catch up with public players. We find that this policy results in 1087 MW less capacity by the end of the sample period, relative to the factual scenario; nor does it incentivize firms to invest earlier, although period-1 investment has become more attractive relative to the factual.

This suggests that per-period or aggregate investment need not increase if new investments in subsequent periods are guaranteed higher access to domestic coal. Investors

consider both the level of domestic coal access guaranteed, as well as the difference between those guarantees across periods. Making one period more appealing relative to the other two could bring in more investment than if all three provide equal coal access, because investors understand that the favorable opportunity to invest is temporary.

The welfare effects of v2 work through the same channel. Production again shifts from public plants to private plants with lower transportation costs and heat rates, and productive efficiency, net of the subsidy, rises by INR 7.4 billion a year (INR 5.8 billion from transportation costs and 1.7 billion from heat rates). Because v2 lowers private investment, unmet peak demand is slightly higher than under the factual policy later in the decade, which costs INR 0.7 billion a year. The net welfare gain is INR 6.7 billion a year.

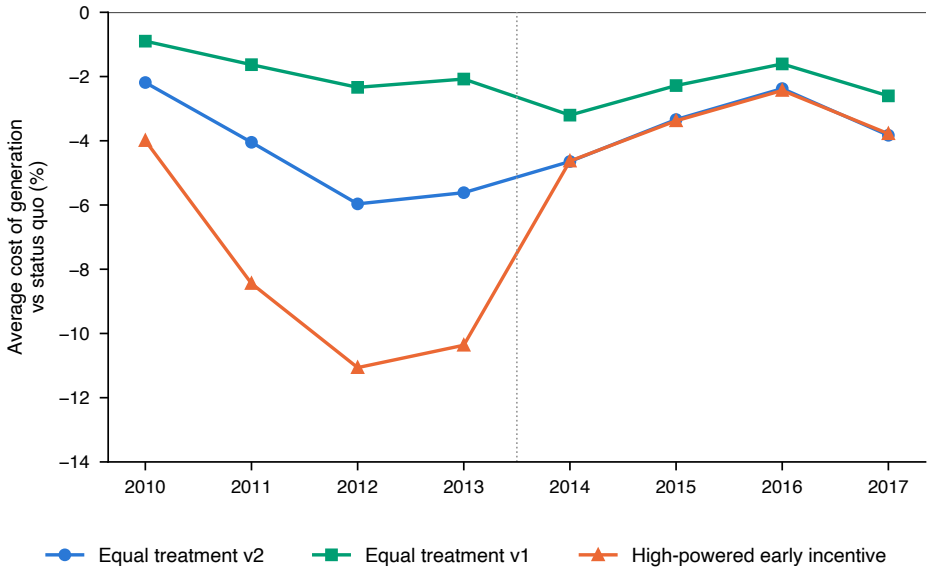
Our second set of counterfactuals is motivated by the observation that forward-looking firms respond to differences in domestic coal access across periods. Regulators could potentially take advantage of this response to alleviate electricity shortages. In period 1, around 15 percent of total demand could not be met in peak hours, even when all available generators produced. Poor access to domestic coal might have prevented potential private investment that could have expanded supply earlier. To encourage early investment, we consider a scenario where regulators can commit ahead of time to a policy path that provides period-1 investment with full access to domestic coal, and denies subsequent investment this benefit.

We find that offering maximal incentives to invest early would raise period-1 investment by more than 9000 MW, at the cost of crowding out investment that would have occurred in subsequent periods under the factual scenario. Overall, capacity at the end of the sample period is similar to that achieved under the factual scenario. However, there are substantial benefits to shifting investment towards earlier periods because generation capacity fell short of meeting demand in those periods.

The welfare effects of the high-powered incentive work through a different channel. Under our timing assumption, it adds about 6 GW of private capacity in 2010–2013, when the supply curve was most often exhausted before demand was met. Unmet peak demand falls by about one percentage point a year through 2014, and the gap narrows as shortages ease later in the decade (Figure 8). The fall in average wholesale electricity cost is also concentrated in 2011–2013, at up to 11 percent (Figure 9). Net of the subsidy, however, productive efficiency rises by only INR 9.4 billion a

year. When demand exceeds supply, every available generator already runs, so the new capacity serves demand that would otherwise go unmet rather than replacing costlier generators. Its value lies instead in lower shortages: unmet energy falls by about 10 TWh a year, worth INR 26.3 billion a year. The net welfare gain, INR 35.7 billion a year, is three to five times that of equalizing domestic coal access, even though capacity at the end of the period is similar under all three policies. Accelerating investment matters because capacity is most valuable when it arrives during a shortage.

Figure 9. Average cost of generation relative to the factual policy



Notes: Percentage change from the factual simulation in the generation-weighted cost of coal and gas, at the prices generators pay, all four markets combined. The change therefore includes the coal subsidy. 2017 covers January to September.

Table 5. Welfare implications of counterfactuals

	Equal treatment v2	Equal treatment v1	High-powered early incentive
Productive efficiency	89.0	47.9	134.3
Heat rate	1.7	1.4	4.4
Transport	5.8	9.0	0.3
Gas displaced	0.0	0.0	4.7
Coal at the subsidised price	81.6	37.6	124.9
Lower shortages	-0.7	0.2	26.3
Subsidy cost	-81.6	-37.6	-124.9
On factual private output, S	-87.7	-45.1	-108.0
On extra private output, X	-10.6	-4.8	-53.9
Domestic coal released and sold, E	16.7	12.3	37.1
Net welfare change	6.7	10.6	35.7

Notes: Entries report average annual welfare changes relative to factual policy, in billions of Indian rupees per year over 2010–2017. Subcomponents may not sum exactly to totals because of rounding.

6 Appendix

Figure 10. Plant Heat Rate by Ownership and Commissioning Year

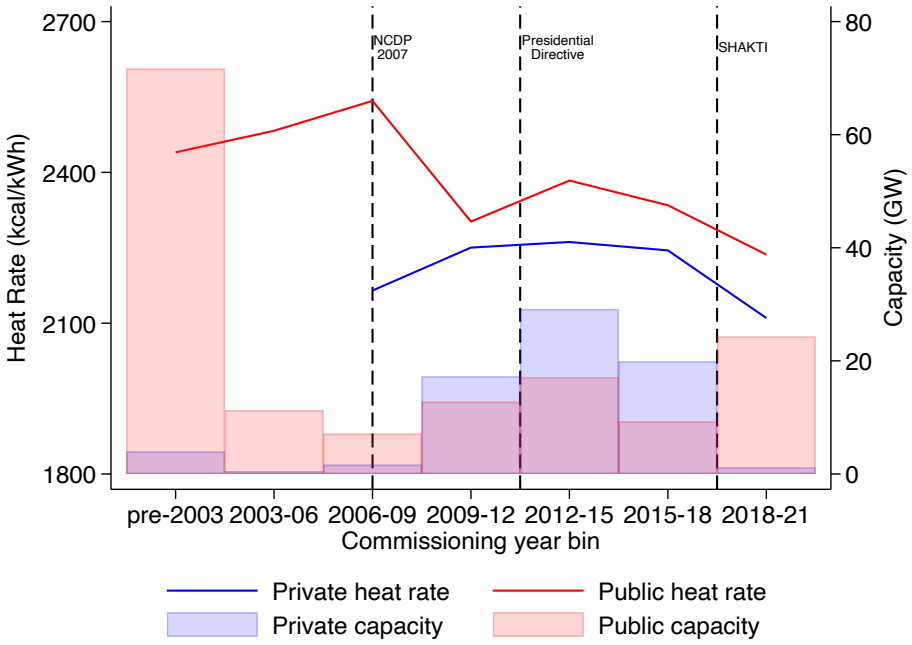


Figure shows plant heat rates (in kcal/kWh) by ownership and vintage. The graph plots heat rate against plant commissioning year for public (red) and private (blue) generators. Lines show average heat rate by ownership while bars show installed capacity (GW, right axis) in each vintage bin. Vertical dashed lines mark key policy changes: New Coal Distribution Policy (2007), Presidential Directive on fuel supply agreements (2012), and SHAKTI policy (2017), explained in detail in section 2.

Figure 11. Counterfactual results by market: costs

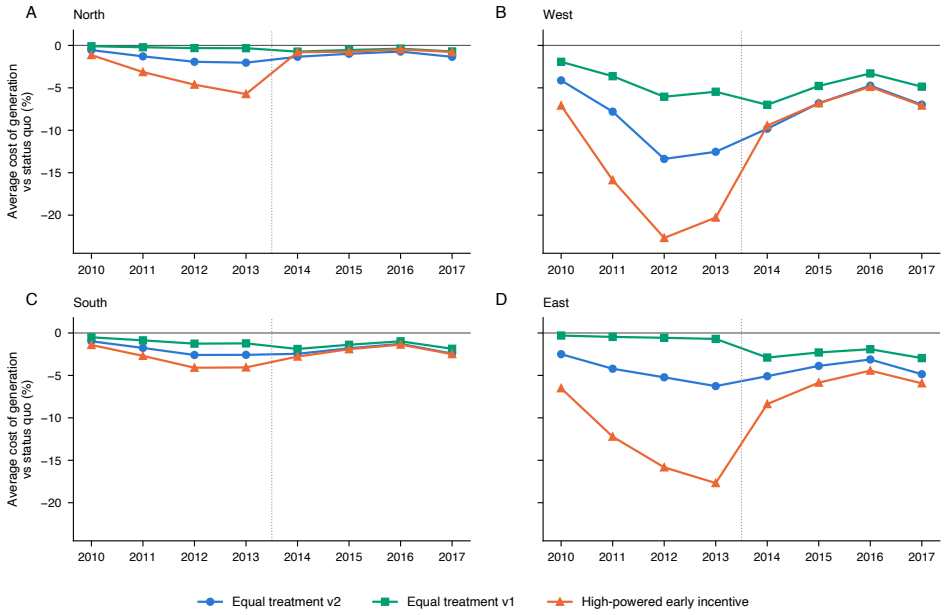
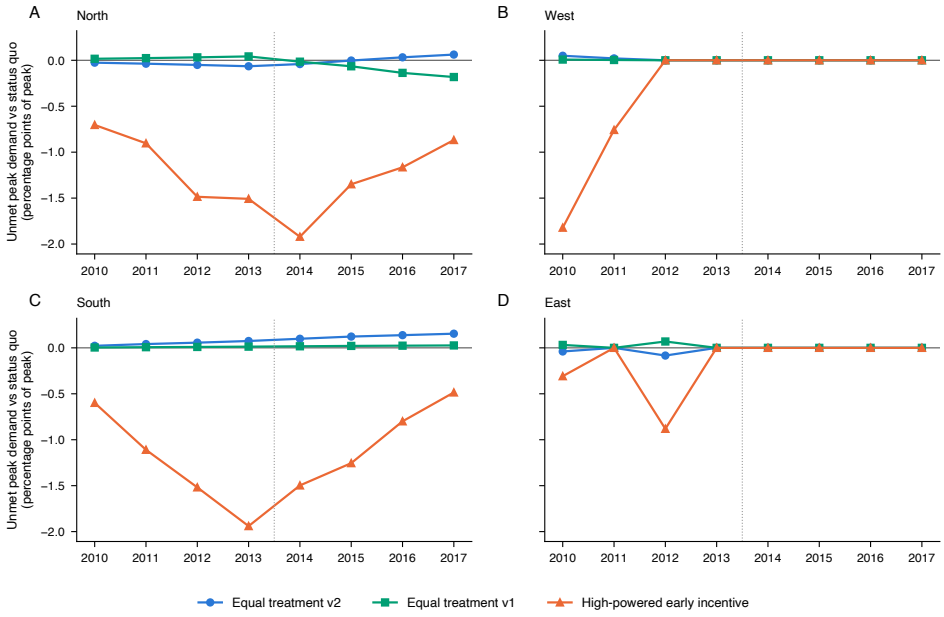


Figure 12. Counterfactual results by market: shortage



References

Allcott, Hunt, Allan Collard-Wexler, and Stephen D. O'Connell, "How Do Electricity Shortages Affect Industry? Evidence from India," *American Economic Review*, March 2016, *106* (3), 587–624.

Asker, John, Allan Collard-Wexler, and Jan De Loecker, "Dynamic Inputs and Resource (Mis)Allocation," *Journal of Political Economy*, 2014, *122* (5), 1013–1063. Publisher: The University of Chicago Press.

Banerjee, Abhijit V. and Esther Duflo, "Do Firms Want to Borrow More? Testing Credit Constraints Using a Directed Lending Program," *The Review of Economic Studies*, April 2014, *81* (2), 572–607.

Besley, Timothy and Maitreesh Ghatak, "Chapter 68 - Property Rights and Economic Development*," in Dani Rodrik and Mark Rosenzweig, eds., *Handbook of Development Economics*, Vol. 5 of Handbooks in Economics, Elsevier, January 2010, pp. 4525–4595.

Busso, Matias, Lucia Madrigal, and Carmen Pages, "Productivity and resource misallocation in Latin America1)," *The B.E. Journal of Macroeconomics*, 2013, *13* (1), 903–932. Publisher: De Gruyter.

Carbon Tracker Initiative, "India Power Tracker: Hourly All-India Generation by Source," https://carbontracker.in/ 2026. Accessed October 2026.

Central Electricity Authority, "Electricity Tariff and Duty and Average Rates of Electricity Supply in India," Technical Report, Ministry of Power, Government of India, New Delhi 2015.

_ , "Monthly Coal Report," Technical Report, Central Electricity Authority 2018-2023. Monthly Reports.

Central Electricity Regulatory Commission, "Report on Short-term Power Market in India: 2014–15," Technical Report, Market Monitoring Cell, Central Electricity Regulatory Commission, New Delhi 2015.

Collard-Wexler, Allan and Jan De Loecker, "Reallocation and Technology: Evidence from the US Steel Industry," *The American Economic Review*, 2015, *105* (1), 131–171. Publisher: American Economic Association.

Comptroller and Auditor General of India, "Performance Audit Report on Allocation of Coal Blocks and Augmentation of Coal Production," Performance Audit Report 7 2012.

Da-Rocha, José-María, Marina Mendes Tavares, and Diego Restuccia, "Firing Costs, Misallocation, and Aggregate Productivity," December 2016.

Elliott, Jonathan T., "Investment, Emissions, and Reliability in Electricity Markets," 2025. Accepted, American Economic Review.

Ericson, Richard and Ariel Pakes, "Markov-Perfect Industry Dynamics: A Framework for Empirical Work," *The Review of Economic Studies*, January 1995, *62* (1), 53–82.

Gowrisankaran, Gautam and Philipp Schmidt-Dengler, "A Computable Dynamic Oligopoly Model of Capacity Investment," March 2025.

_ **, Ashley Langer, and Wendan Zhang**, "Policy Uncertainty in the Market for Coal Electricity: The Case of Air Toxics Standards," *Journal of Political Economy*, June 2025, *133* (6), 1757–1795.

Hsieh, Chang-Tai and Peter J. Klenow, "Misallocation and Manufacturing TFP in China and India*," *The Quarterly Journal of Economics*, November 2009, *124* (4), 1403–1448.

Ifrach, Bar and Gabriel Y. Weintraub, "A Framework for Dynamic Oligopoly in Concentrated Industries," *The Review of Economic Studies*, 2017, *84* (3), 1106–1150.

Kumar, Alok and Sushanta Chatterjee, *Electricity Sector in India: Policy and Regulation*, Oxford University Press, October 2012.

Mathavan, Deeptha, "From Dabhol to Ratnagiri: The electricity act of 2003 and reform of India's power sector," *Colum. J. Transnat'l L.*, 2008, *47*, 387.

Melitz, Marc J., "The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity," *Econometrica*, 2003, *71* (6), 1695–1725. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1468-0262.00467.

_ **and Stephen J. Redding**, "Heterogeneous Firms and Trade," *Handbook of International Economics*, 2014, *4*, 1–54. Publisher: Elsevier.

Midrigan, Virgiliu and Daniel Yi Xu, "Finance and Misallocation: Evidence from Plant-Level Data," *American Economic Review*, February 2014, *104* (2), 422–458.

Ministry of Coal, October 2007. New Coal Distribution Policy.

_ , "Annual Report," Technical Report, Ministry of Coal 2014.

_ , "SHAKTI (Scheme for Harnessing and Allocating Koyala Transparently in India) Policy," Policy Notification 23011/15/2016-CPD/CLD, Government of India May 2017.

Ministry of Power, "National Tariff Policy, Resolution No. 23/2/2005 (III)," *Gazette of India*, 2006, pp. 389, 415.

Olley, G. Steven and Ariel Pakes, "The Dynamics of Productivity in the Telecommunications Equipment Industry," *Econometrica*, 1996, *64* (6), 1263–1297. Publisher: [Wiley, Econometric Society].

Pakes, Ariel and Paul McGuire, "Computing Markov-Perfect Nash Equilibria: Numerical Implications of a Dynamic Differentiated Product Model," *The RAND Journal of Economics*, 1994, *25* (4), 555–589.

Peters, Michael, "Heterogeneous Markups, Growth, and Endogenous Misallocation," *Econometrica*, 2020, *88* (5), 2037–2073. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA15565.

Restuccia, Diego and Richard Rogerson, "The Causes and Costs of Misallocation," *Journal of Economic Perspectives*, August 2017, *31* (3), 151–174.

Ryan, Nicholas, "Contract Enforcement and Productive Efficiency: Evidence From the Bidding and Renegotiation of Power Contracts in India," *Econometrica*, 2020, *88* (2), 383–424. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA17041.

_ , "The Competitive Effects of Transmission Infrastructure in the Indian Electricity Market," *American Economic Journal: Microeconomics*, May 2021, *13* (2), 202–242.

Sehgal, Anurag and Rahul Tongia, "Coal Pricing and Allocations in India," in Rahul Tongia, Anurag Sehgal, and Puneet Kamboj, eds., *Future of Coal in India*, Brookings India, 2020.

Standing Committee on Energy, "Stressed/Non-performing Assets in Electricity Sector," Thirty-Seventh Report 2017-18 (Sixteenth Lok Sabha), Ministry of Power, Government of India March 2018. Presented to Lok Sabha on 07.03.2018, Laid in Rajya Sabha on 07.03.2018.

Taneja, Mansi, "Indonesia's coal pricing norms hit Tata's Mundra UMPP," *Business Standard*, Jan 2013.

Weintraub, Gabriel Y., C. Lanier Benkard, and Benjamin Van Roy, "Markov Perfect Industry Dynamics with Many Firms," *Econometrica*, 2008, *76* (6), 1375–1411.

Zwick, Eric and James Mahon, "Tax Policy and Heterogeneous Investment Behavior," *American Economic Review*, 2017, *107* (1), 217–248.